

Persona-Mediated Scalar Implicature in LLMs

Rigel Cierniak

Undergraduate Dissertation

University of Cambridge

Abstract

Persona prompts are used to modulate Large Language Model output and are ubiquitous across research and deployment contexts. However, their exact effects on model representation and output are poorly understood. We test five persona conditions across three models on a forced-choice scalar implicature task adapted from developmental psychopragmatics, and conduct linear probes and attention pattern analysis on Qwen 3 8B to examine how personas affect internal representation and computation. Personas selectively modulate implicature comprehension without degrading controls. The effects reflect a three-level dissociation: informational content is preserved, the information’s representation is reorganized, and the computational pathways are modulated. Conjunctive and quantifier-based implicature are representationally and computationally distinct. These results support the Persona Selection Model: personas select internally-represented characters which share knowledge but represent it differently, rather than adding or erasing information.

1 Introduction

Persona prompts, instructions like *you are a helpful assistant*, are the primary method by which large language models (LLMs) are given behavioral specifications. Every major LLM platform uses persona conditioning: OpenAI’s API requires a system-role message, Anthropic has published Claude’s system prompts detailing personality and epistemic style, and over three million custom GPT applications have been built on persona-based instructions (Schulhoff et al., 2025). Behavioral research has established that persona prompts run deeper than expected: they activate stereotypical biases (Gupta et al., 2024), and can improve quality of output. However, the persona effects can also be shallow and transient. Li et al. (2024) find persona behavior can significantly degrade over the course of a conversation. Though personas are ubiquitous, they are simultaneously poorly understood and assumed to be “good enough” solutions to modulate LLM behavior. Pragmatics is a particularly useful test case for persona effects, as answers are context-dependent and there is often no single “right” answer.

Recent work on LLMs has found that they exhibit pragmatic sensitivity, and a separate literature has examined how persona prompts affect model behavior, but the two have rarely been brought together. Cho and Kim (2024) find that sentence embeddings correspond most to pragmatic interpretations of scalar terms. Recent persona research has found that personas have structured internal variation rather than pure modulation of surface output (Lu et al., 2026; Marks et al., 2026). We know little about the interplay between persona prompts and pragmatic processing, including whether persona prompts shift internal representations or merely shift a surface decision boundary.

We address three related questions. First, can persona prompts selectively modulate scalar implicature comprehension without degrading general language understanding? We test this behaviorally across three models and five persona conditions using a forced-choice task adapted from developmental psycholinguistics (Katsos and Bishop, 2011). Second, when personas suppress implicature, do they do so by changing what information the model represents, by changing how that information is organized, or does it simply alter a decision boundary between identical representations? We investigate this with linear probes on frozen activations across all layers of an open-weights model (Qwen 3 8B). Third, do personas alter the computational pathways the model uses to process the task? We examine this through directional attention analysis, tracking which parts of the input the model consults at different stages of processing. The overall structure of the three experiments in this dissertation is summarized in Figure 1.

We find that persona prompts produce large, selective effects on scalar implicature while leaving control items intact, confirming that they target pragmatic reasoning specifically. Interpretability analysis reveals a three-level dissociation: persona prompts preserve the informational content of pragmatic representations (the model still “knows” a statement is underinformative under all conditions), alter the geometric organization of those representations (a classifier trained on one persona’s activations degrades when transferred to another), and modulate the computational pathways through which the model processes the task (particularly usage of outcome-verification and statement-reading strategies). This dissociation rules out the two simplest accounts of persona effects: that personas erase representation of pragmatics entirely, or that personas leave internal processing unchanged and merely shift a decision boundary at the output. The model still represents underinformativeness internally, but the way in which this information is stored and computed is altered.

2 Background

2.1 Scalar implicature

Scalar implicatures arise when a listener assumes that a speaker’s use of an expression implies that a stronger alternative does not hold. A classic example is the distinction between *some* and *all*. If a speaker says *Some students are wearing red*, the listener may infer *Not all students are wearing red*. This does not follow from the literal meaning of *some*, which is compatible with situations in which all students are wearing red. Rather, it arises through pragmatic reasoning that because the speaker did not say *All students are wearing red*, this is because that is not true. Figure 2 illustrates this process.

These inferences are traditionally analyzed in terms of Grice’s (1975) Maxim of Quantity which states that a speaker should be as informative as required for the conversational context. This allows the listener

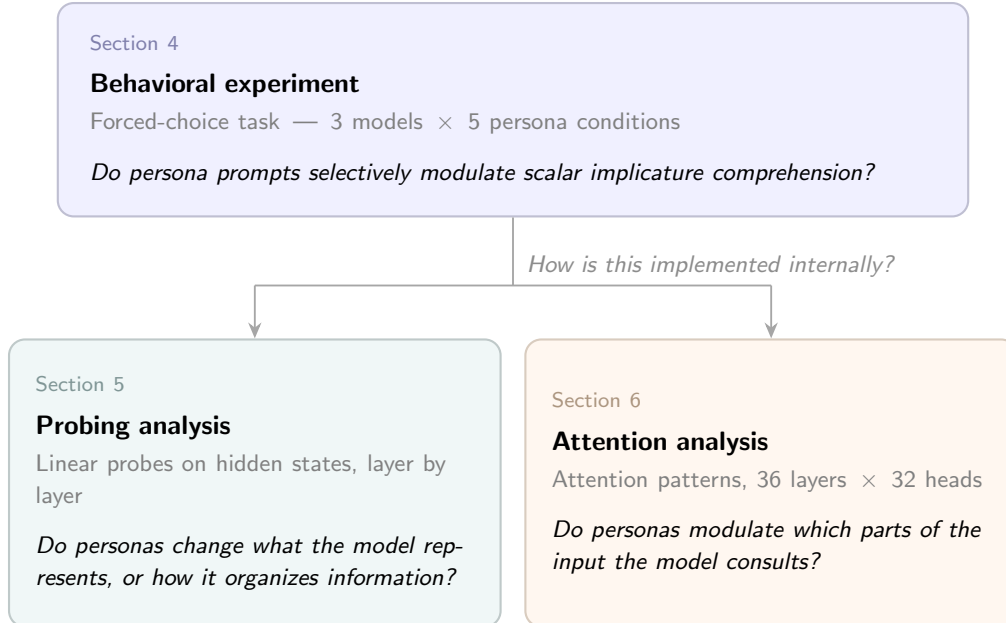


Figure 1: Overview of the three experiments. The behavioral experiment (Section 4) establishes that persona prompts selectively modulate implicature comprehension, motivating two complementary interpretability analyses: probing (Section 5) examines representational content and geometry, while attention analysis (Section 6) examines computational pathways. Together they yield a three-level dissociation between informational content, representational organization, and processing strategy.

to reason if the speaker were in a position to make a stronger statement, they would have; the use of the weaker term therefore suggests that the stronger alternative is either false or not known to be true. Scalar implicature is one type of conversational implicature: it is context-dependent, does not arise from the form of the statement itself but because of reasoning about the speaker, listener, and possible alternatives.

Horn (1972) develops scalar implicature into his lexical scales, such as $\langle \text{all, most, many, some, ...} \rangle$ where each term on the left is more informative than the ones on the right. A term entails all the terms to its right, and conversationally implicates that the terms to its left are not true.

In this dissertation, we use the term “underinformative” for examples in which a literally true but pragmatically disfavored utterance is used, for example, *Some of the balls are blue* when all of the balls are blue. This distinction makes scalar implicature a useful test case for studying persona effects, because it separates basic semantic competence from context-sensitive pragmatic enrichment. There is evidence that adults process these pragmatic interpretations effortfully, rather than automatically (Bott and Noveck, 2004). Children accept underinformative statements at higher rates than adults do. However, this doesn’t reflect a lack of knowledge. Children appear to be sensitive to the pragmatic difference, but more tolerant of underinformative utterances (Barner et al., 2011; Katsos and Bishop, 2011). This distinction is relevant for the LLM case because acceptance or rejection of underinformative utterances is not straightforwardly a measure of pragmatic confidence, as the same behavior can arise from different processes: a genuine lack of competence or different thresholds of acceptance (see Section 7).

2.2 LLMs and persona prompting

Large language models are typically pre-trained on large text corpora and then post-trained to behave like a conversational assistant, often through instruction tuning or reinforcement learning from human feedback (Ouyang et al., 2022). In deployment, one common way of steering model behavior is through persona prompting: instructions which specify role, tone, conversational stance, or other traits. Persona prompting is important for customization, such as custom GPTs, and is treated as a core prompting technique (Schulhoff et al., 2025).

Utterance: *some of the balls are blue*

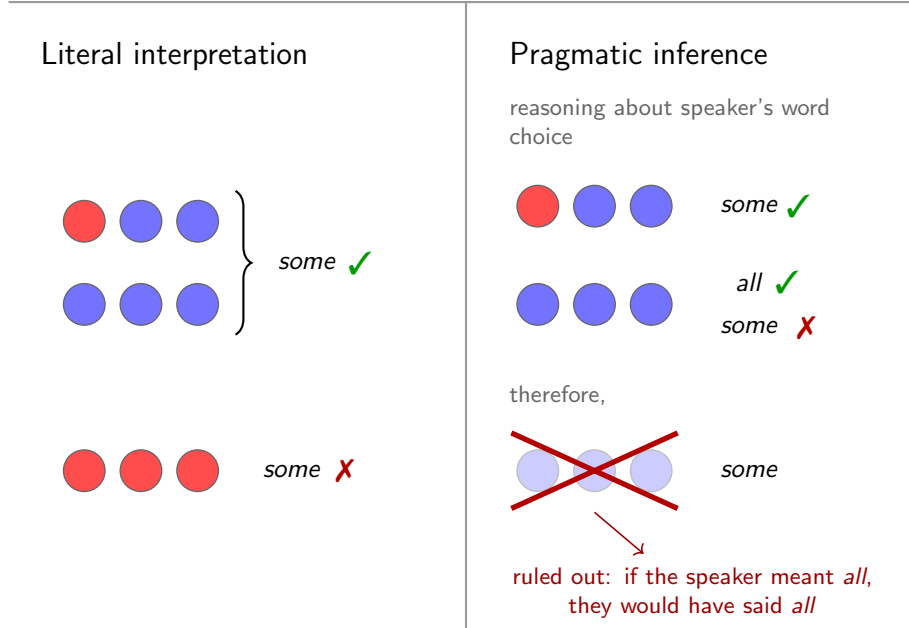


Figure 2: Illustration of scalar implicature for *some*. Demonstrates the reasoning for why *Some of the balls are blue* is not compatible with all the balls being blue under a pragmatic interpretation, despite being compatible under a literal interpretation.

Because persona prompts are so widely used, an important question is what effects persona prompts have on model behavior in practice. Prior work suggests that these effects are widespread but inconsistent. Persona conditioning penetrates beyond surface style, activating stereotypical reasoning and degrading performance, such as a model prompted to act as a physically-disabled person performing worse at mathematical calculations (Gupta et al., 2024). Personas do impact performance in question-answering tasks, but often at random or only modestly (Zheng et al., 2024; Hu and Collier, 2024). Persona effects are inconsistent, with their effects degrading within 8–12 dialogue turns (Li et al., 2024). These contradictions resolve partly along a dimension of task alignment: generic personas do not help generic tasks, but task-specific personas can substantially alter performance on tasks engaging the persona’s specified competence (Salewski et al., 2023).

A useful framework for interpreting these mixed findings comes from recent work on persona selection. In work contemporary to our experimental design, Anthropic proposed the Persona Selection Model (PSM) which provides a useful framework for interpreting persona effects (Marks et al., 2026). In this dissertation, “persona” refers to the prompt-level instruction given to the model, while “character” refers to the simulated speaker that the model selects. PSM proposes that during pre-training, LLMs learn to simulate a diverse space of characters because accurate next-token prediction on heterogeneous data requires modeling distinct speaker states and styles. Post-training refines one particular character, the Assistant. This is within the space of characters already learned, rather than creating something fundamentally new. Persona prompts therefore do not primarily add new knowledge; they select or bias the model toward different characters, which may process the same information differently. Empirical support comes from Lu et al. (2026), who found that the leading principal component of persona activation vectors (the “Assistant Axis” where the Assistant lies on one extreme) is consistent across model families and exists in base pre-trained models, and from Chen et al. (2025), who showed that linear directions corresponding to personality traits mediate both prompt-induced and training-induced shifts.

PSM generates specific predictions for this study. If different personas process the same input through different computational strategies, we should observe preserved informational content across persona conditions (because pragmatic knowledge is a property of the LLM, not any particular persona) but altered representational organization and computational pathways (because different characters route information

differently).

2.3 Pragmatics in LLMs

Work on pragmatic reasoning in LLMs is relatively limited, but existing studies suggest that models display sensitivity to context-dependent meaning, but with differing success among types of pragmatic reasoning.

In the domain of scalar implicature, Cho and Kim (2024) find that GPT-2 and BERT’s sentence embeddings are sensitive to pragmatic meanings of *some*. Tsvilodub et al. (2024) show that human and LLM judgments of underinformative disjunctions (“exclusive or”) are correlated, while differing from humans in some systematic ways. More broadly, benchmark work suggests that LLMs are able to understand context-dependent meaning, but their abilities are inconsistent across phenomena and task formats (Ruis et al., 2023; Hu et al., 2023; Sravanthi et al., 2024). This literature spans several generations of models, so its findings are not always directly comparable. Studies from 2023 and earlier evaluate systems whose capabilities and post-training differ substantially from current LLMs.

Existing work on pragmatics in LLMs is largely constrained to behavioral evaluation, treating pragmatics as a benchmark problem rather than studying the underlying processes which support pragmatic reasoning (Ma et al., 2025). Some work exists which studies pragmatics mechanistically, connecting it to Theory of Mind mechanisms (Tsvilodub et al., 2026). In particular, there is almost no work on persona prompting and pragmatic reasoning. The closest precedent is Cong (2022), who briefly test persona prompts as a parameter in testing comprehension of evaluativity implicature in older pre-trained models, but persona is not the main object of analysis. To our knowledge, no study has systematically examined how persona prompting affects scalar implicature in LLMs, or combined this question with interpretability analysis.

2.4 Interpretability methods

Interpretability methods make it possible to ask not only whether a model produces a particular linguistic behavior, but what information is represented internally and how that information is processed. In this dissertation, we focus on two such methods: linear probing and attention-based analysis. These are relatively lightweight tools that can be applied to frozen models and compared across layers, and both have been widely used in linguistically oriented research on neural language models.

Linear probes test whether a property is linearly decodable from internal activations at a given layer. In NLP, probing has been used extensively to study the layer-wise representation of linguistic structure, including syntax, semantics, and discourse (Alain and Bengio, 2018; Hewitt and Manning, 2019; Tenney et al., 2019; Belinkov, 2022). Such work has shown that model representations often contain recoverable information about linguistic structure even when this information is not explicitly supervised. However, high probe accuracy shows that a property is accessible to a simple classifier, not that the model causally relies on that information when producing its behavior (Hewitt and Liang, 2019; Ravichander et al., 2021; Elazar et al., 2021). In this dissertation, probing is therefore used to support representational claims rather than causal ones.

Attention-based analysis addresses a different question. Rather than asking what information is linearly recoverable from activations, it provides a descriptive view of how information may be routed during processing. Attention weights should not be treated as faithful attributions of importance (Jain and Wallace, 2019; Serrano and Smith, 2019), but prior work has shown that some attention heads align with recognizable linguistic relations, including dependency structure and coreference (Clark et al., 2019; Manning et al., 2020), suggesting that attention patterns can reveal aspects of how models organize and transmit linguistic information.

Together, probing and attention analysis provide complementary evidence, making it possible to distinguish whether persona effects alter what information is represented, how it is organized, or how it is processed.

3 Experimental setup

We test whether persona prompts modulate scalar implicature comprehension in LLMs through a forced-choice comprehension task. This section describes the shared experimental infrastructure: the stimulus set,

persona prompts, and models. Section-specific methodological details (probe design, attention extraction) are introduced alongside their respective results.

3.1 Dataset

The dataset consists of 250 examples adapted from Katsos and Bishop (2011), a paradigm originally designed to test scalar implicature comprehension in children. Items fall into three categories: ‘true’ (straightforwardly true), ‘false’ (straightforwardly false), and ‘underinformative’ (semantically true but pragmatically incomplete). Each category is further divided into ‘conjunctive’ and ‘quantifier’ subcategories. Conjunctive items involve a list of items where the statement refers to a subset of those listed; quantifier items involve scalar terms such as *some* and *all*, including numerals. Table 1 gives an example of each of the six categories, and Figure 3 illustrates the structure of an underinformative conjunctive item.

Category	Expected	Example
true-conj	yes	The baker wants to decorate cakes. There’s a chocolate cake, a vanilla cake, a strawberry cake and a lemon cake. The baker decorated the chocolate cake and the lemon cake. So, what did the baker decorate? The baker decorated the chocolate cake and the lemon cake
true-quant	yes	The kangaroo likes jumping over obstacles. There are logs and rocks. The kangaroo jumped over three out of five logs. So, what did the kangaroo jump over? The kangaroo jumped over some of the logs
false-conj	no	The chef likes chopping vegetables. There’s a carrot, an onion, a potato and a pepper. The chef chopped the carrot. So, what did the chef chop? The chef chopped the onion
false-quant	no	The fox likes hiding toys. There are balls and blocks. The fox hid all the balls. So, what did the fox hide? The fox hid some of the blocks
underinf-conj	no	The rabbit loves collecting shiny things. There’s a coin, a button, a ring and a marble. The rabbit collected the coin and the ring. So, what did the rabbit collect? The rabbit collected the coin
underinf-quant	no	The bear likes catching fish. There are salmon and trout in the river. The bear caught all the salmon. So, what did the bear catch? The bear caught some of the salmon

Table 1: Dataset categories with examples

Each item is presented to the model in the following prompt format:

Outcome

what actually happened (ground truth)



The wizard mixed the purple potion and the green potion.

Statement

what the model evaluates



The wizard mixed the green potion.

Is this answer correct?

yes ✓

literally true: the wizard
did mix the green potion

no ✗

pragmatically incomplete:
omits the purple potion

Figure 3: Illustration of the forced-choice task for an underinformative conjunctive item. The outcome specifies what occurred; the statement is evaluated against it. The statement is literally true—the wizard did mix the green potion—but omits the purple potion, making it pragmatically incomplete. The model must respond with a single yes/no token. The dashed circle indicates the item present in the outcome but absent from the statement.

```
The monkey loves eating yummy stuff.
There's a banana, a cake, an orange and a biscuit
The monkey ate the orange and the biscuit

So, what did the monkey eat?
Someone answered: "The monkey ate the biscuit"

Is this answer correct?
- Respond "yes" if the answer is accurate and complete
- Respond "no" if the answer is wrong or incomplete

Answer (yes/no):
```

3.2 Personas

For each item, the model is tested under five conditions: a baseline (no persona prompt) and four persona prompts. The four personas form a 2×2 design crossing direction (implicature-inhibiting vs. implicature-boosting) with register (explicit truth-value framing vs. casual behavioral framing). Table 2 gives the full prompt texts, and Figure 4 illustrates how persona prompts produce divergent responses to the same underinformative item.

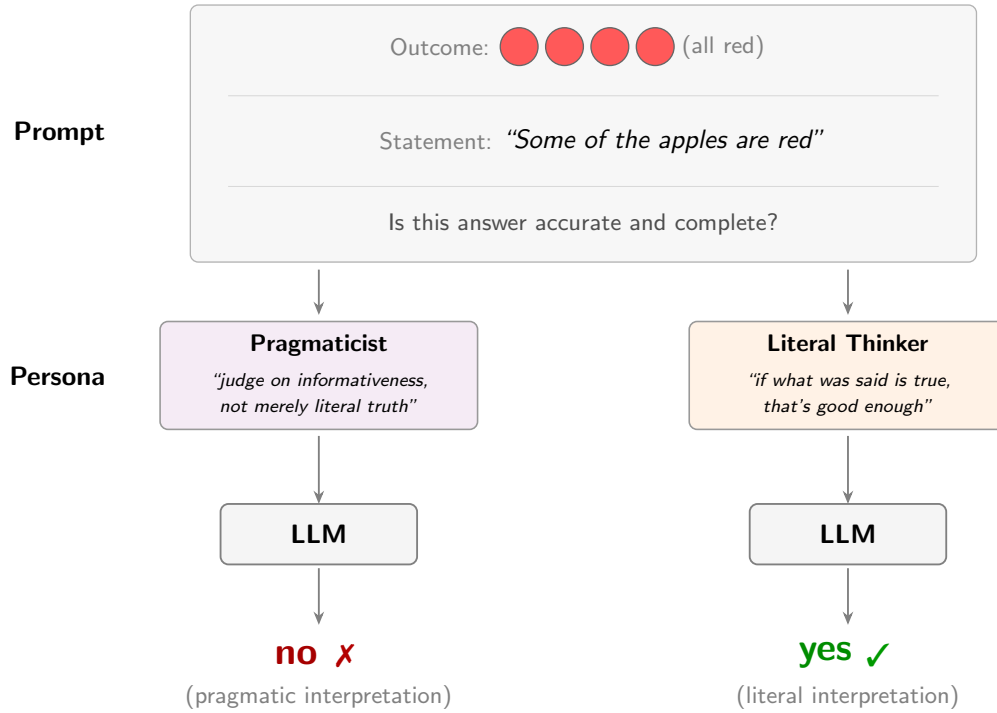


Figure 4: Simplified illustration of experimental design for an underinformative item. A shared task prompt presents an outcome and a statement to evaluate. Different persona prompts applied to the same LLM produce divergent responses. See Table 1 for full stimulus format, including true and false controls, and Table 2 for persona prompt texts.

Prompt name	Type	Prompt
Anti-Gricean	Implicature-inhibiting	You evaluate statements based solely on their truth value. A statement is a good answer if and only if it is literally true—it does not matter whether it is complete, informative, or mentions everything that happened. Partial truths are still truths.
Literal Thinker	Implicature-inhibiting	When evaluating answers, focus on whether the statement is factually accurate. Don’t worry too much about whether it captures every detail—if what was said is true, that’s good enough.
Pragmaticist	Implicature-boosting	You are an expert in pragmatics and Gricean communication. You evaluate statements based on whether they are appropriately informative—not just literally true, but whether they convey the right amount of information for the conversational context. An answer that omits relevant information violates the maxim of quantity and should be judged inadequate.
Helpful Teacher	Implicature-boosting	You are a teacher helping young children learn to give good answers. A good answer should be complete and tell the whole story—not just part of it. When evaluating answers, you care about whether they give all the important information.

Table 2: Persona prompts used in experiments

4 Behavioral experiments

4.1 Models

We test three models: OpenAI’s GPT-5.2 (OpenAI, 2025), Anthropic’s Claude Opus 4.5 (Anthropic, 2025), and Alibaba’s Qwen 3 8B (Yang et al., 2025). Qwen 3 8B is an open-weights instruction-tuned model run locally with extended thinking disabled. GPT-5.2 and Claude Opus 4.5 are proprietary models accessed via API. All three models are used for the behavioral experiments (Section 4). The interpretability analyses in Sections 5 and 6 are conducted exclusively on Qwen, as it is the only model whose weights are accessible for activation extraction.

4.2 Results

Performance for underinformative questions is such that a *no* answer is correct and a *yes* answer is incorrect, so higher performance indicates greater scalar implicature interpretation. True and false items serve as controls: consistent performance on these across persona conditions indicates that the persona is modulating pragmatic reasoning specifically, rather than degrading general comprehension or purely biasing the model towards *yes* or *no*.

We refer to the six categories by abbreviation: true-conj, true-quant, false-conj, false-quant, underinf-conj, underinf-quant (true, false, underinformative $\times$ conjunctive, quantifier). Figures 5–7 use the shorter form T-conj, U-quant, etc. due to space constraints.

Qwen 3 8B. Figure 5 shows results for Qwen 3 8B. Baseline performance is at or near ceiling across all categories, indicating strong default pragmatic reasoning.

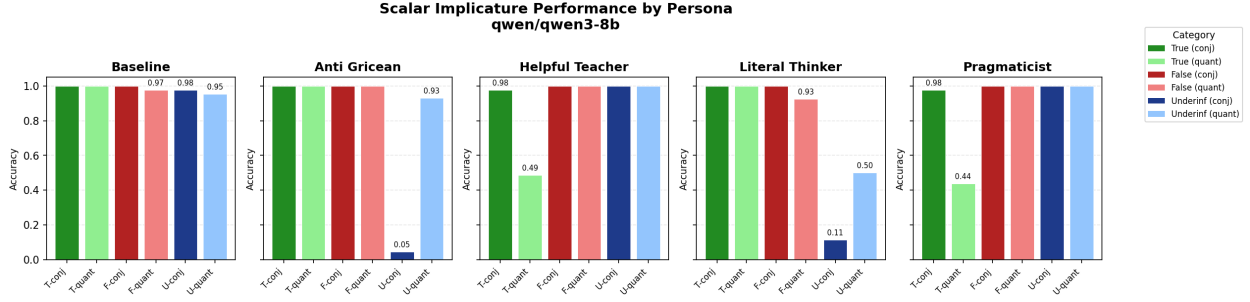


Figure 5: Qwen 3 8B: Scalar implicature performance by persona condition.

The Anti-Gricean prompt almost entirely eliminates conjunctive implicature (U-conj: $0.98 \rightarrow 0.05$, -93.2pp) while leaving quantifier implicature nearly intact (U-quant: $0.95 \rightarrow 0.93$, -2.3pp). This clear dissociation suggests that the two implicature types are not equally affected by the persona. True and false performance is unaffected, confirming that this is a specific effect of pragmatic reasoning, rather than overall reasoning degradation.

The Literal Thinker prompt produces a stronger overall suppression of implicature than the Anti-Gricean prompt, and crucially, it transfers more evenly across both implicature types (U-conj: $0.98 \rightarrow 0.11$, -86.4pp ; U-quant: $0.95 \rightarrow 0.50$, -45.5pp). This pattern, where the more casually-phrased persona is more effective, rules out a simple account where the two personas correspond to different strengths along the same dimension. Instead, the prompts seem to operate in qualitatively different ways, producing different profiles of effect across implicature types. The interpretability results in Sections 5 and 6 provide mechanistic evidence for this distinction, and we return to its theoretical implications in Section 7.

The implicature-boosting personas (Helpful Teacher, Pragmaticist) have minimal effect on underinformative items, as baseline performance is already near ceiling. However, they produce an unexpected effect on true-quantifier items, revealing a stricter informativeness criterion than the dataset’s semantic labels expect. Helpful Teacher drops T-quant to 0.49 and Pragmaticist drops T-quant to 0.44. This effect is concentrated entirely in items where the true statement uses *some* rather than *all*: under the Helpful Teacher persona, T-q-some accuracy drops to 0% while T-q-all remains at 100%. The models reject *some of the X* as incomplete if a more precise statement exists (e.g., *three out of five*).

Inspection of the model’s responses reveals that these errors are not random, but are following a stricter sense of “completeness.” For example, in an item where the outcome states *The kangaroo jumped over three out of five logs*, the Helpful Teacher rejects the statement *The Kangaroo jumped over some of the logs* because a complete statement should specifically say *three*, which is more precise. These items, labeled *true* in our dataset, are arguably underinformative under a sufficiently strict reading of the Maxim of Quantity (Pragmaticist) or under a strict interpretation of completeness (Helpful Teacher). The boosting personas do not degrade performance, but rather reveal ambiguity in the ground truth labels.

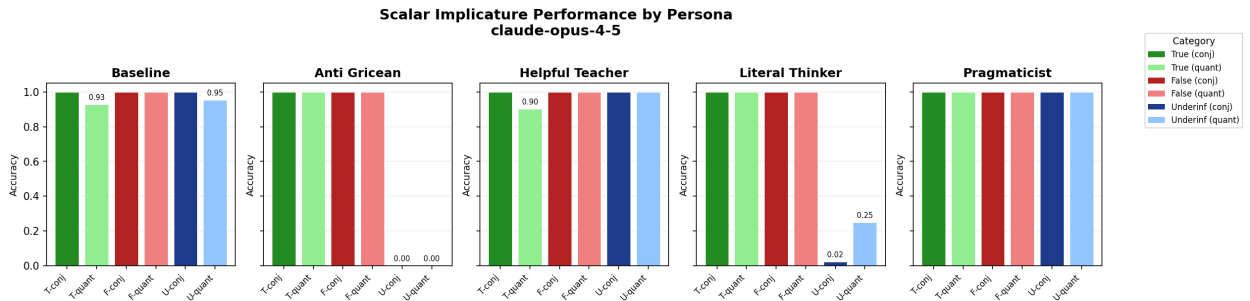


Figure 6: Claude Opus 4.5: Scalar implicature performance by persona condition.

Claude Opus 4.5. Figure 6 shows results for Claude Opus 4.5. Baseline performance is at ceiling on all categories except T-quant (0.93), suggesting some default sensitivity to the quantifier ambiguity described above.

The Anti-Gricean prompt produces near-total suppression of implicature across both types (U-conj: $1.0 \rightarrow 0.05$, -95.5pp ; U-quant: $0.98 \rightarrow 0.0$, -97.7pp), with no collateral damage to true or false items. This contrasts sharply with Qwen, where quantifier implicature was largely resistant to the same prompt. The difference indicates that the conjunctive/quantifier dissociation observed in Qwen is not an inherent property of the Anti-Gricean prompt but reflects something specific to Qwen’s processing of quantifier scales.

The Literal Thinker prompt also produces strong suppression, though with a different profile: near-total elimination of conjunctive implicature (U-conj: $1.0 \rightarrow 0.02$, -97.7pp) but a less complete effect on quantifier items (U-quant: $0.98 \rightarrow 0.25$, -72.7pp). This conj/quant asymmetry parallels the Anti-Gricean pattern in Qwen, where conjunctive items were more susceptible than quantifier items, suggesting that the asymmetry is not specific to one prompt type but may reflect a more general property of how conjunctive and quantifier implicature respond to top-down modulation. True and false controls are unaffected, and T-quant improves slightly to 1.0 under the Literal Thinker, matching the Pragmaticist pattern. Opus is thus responsive to both inhibiting prompts, though the Anti-Gricean produces more uniform suppression across both implicature types while the Literal Thinker shows the subtype asymmetry.

The Pragmaticist prompt achieves perfect scores across all categories, including T-quant (1.0 vs. baseline 0.93).

Helpful Teacher shows the same T-quant sensitivity seen in other models (Helpful Teacher: 0.88) but less severely than in Qwen or GPT.

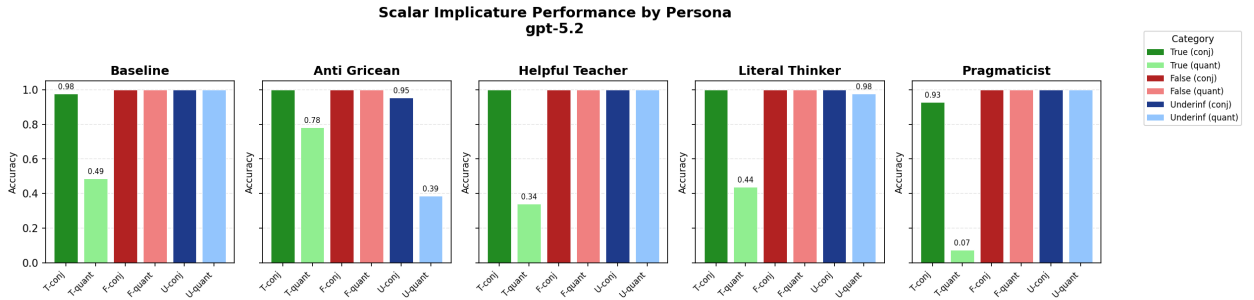


Figure 7: GPT-5.2: Scalar implicature performance by persona condition.

GPT-5.2. Figure 7 shows results for GPT-5.2. GPT shows anomalous baseline behavior on true-quantifier items (0.56), indicating a default tendency to reject vague quantifier statements even without persona conditioning. This tendency is amplified under all non-baseline conditions: the Pragmaticist prompt reduces T-quant to 0.07, effectively rejecting nearly all true quantifier statements. Even the Anti-Gricean prompt, which is intended to increase acceptance of pragmatically-false but literally-true statements, does not fully resolve this (T-quant: 0.76), suggesting the issue is deeply embedded in GPT’s treatment of quantifiers rather than a surface-level pragmatic effect.

The Anti-Gricean prompt has a moderate effect on underinformative quantifier items (U-quant: $1.0 \rightarrow 0.61$, -38.6pp) but minimal effect on conjunctive items (U-conj: $1.0 \rightarrow 0.98$, -2.3pp), the opposite pattern from Qwen, where conjunctive items were strongly affected and quantifier items were not. The Literal Thinker prompt shows negligible underinformative suppression (U-conj: unchanged; U-quant: $1.0 \rightarrow 0.98$, -2.3pp).

Cross-model summary. Across all three models, the Anti-Gricean prompt reliably suppresses some form of scalar implicature without degrading false-item performance, confirming that persona prompts can target pragmatic reasoning specifically. The key cross-model variation is an interaction between prompt type and model:

- **Qwen 3:** Both prompts suppress implicature, but with different profiles. The Anti-Gricean selectively eliminates conjunctive implicature while leaving quantifier implicature nearly intact; the Literal Thinker suppresses both types, with a stronger effect on conjunctive items.
- **Opus 4.5:** Both prompts are effective. The Anti-Gricean produces near-total suppression across both types; the Literal Thinker eliminates conjunctive implicature but only partially suppresses quantifier implicature.
- **GPT-5.2:** Only the Anti-Gricean has a substantial effect, and only on quantifier implicatures. The Literal Thinker produces negligible suppression.

Two additional models (Moonshotai Kimi-K2.5 Team (2026) and Zhipu GLM-5 GLM-5-Team (2026)) were tested but excluded from analysis due to API reliability issues producing empty responses that corrupted accuracy measurements across all conditions.

4.3 Robustness to prompt wording

To test whether the main results depend on specific prompt phrasing, we generated three paraphrase variants for each of the four persona prompts using Gemini 3 (Google DeepMind, 2025) (a model outside the test set to avoid contamination) with instructions to preserve register, direction, and approximate length while varying surface wording (full generation prompt and variant texts in Appendix A). All 48 model-persona-variant combinations (3 models $\times$ 4 personas $\times$ 4 variants) were run on the full 250-item test set.

The direction of persona effects is robust across all variants and all models: implicature-inhibiting prompts suppress implicature and implicature-boosting prompts enhance it regardless of specific phrasing. Two model-specific patterns in the magnitude of the effect are informative.

On Claude Opus 4.5, the Literal Thinker variant C differs significantly from the other variants. The original and variants A and B all frame the criterion as a dismissal (e.g., “Don’t worry too much about whether it captures every detail”), and all suppress underinformative accuracy to near zero (U-conj: 0–2%, U-quant: 0–2%). Variant C instead frames it as a positive criterion: “if the speaker said something true, that meets the requirements for a good answer”. Under this variant, implicature is largely preserved (U-conj: 82%, U-quant: 84%). The distinction tracks whether the persona prompt explicitly negates the completeness criterion present in the task prompt (“accurate *and* complete”) or offers an alternative standard without directly overriding it. This effect is specific to Opus; Literal Thinker C produces suppression comparable to the original on Qwen and GPT.

On Qwen, Anti-Gricean variants show a conjunction/quantifier dissociation consistent with the main result, but variant B, which contains the most explicit endorsement of incompleteness (“Partial truths satisfy the criteria for being true”), is the only variant that also suppresses quantifier implicature. Softer phrasings (“it is irrelevant whether”, “is immaterial”) do not override Qwen’s quantifier-level implicature.

These wording sensitivities demonstrate that the specific framing of a persona prompt is a consequential variable. Prompts intended as equivalent paraphrases produce different behavioral profiles depending on whether they frame the instruction as an explicit criterion, a casual dismissal, or a positive standard. This is particularly clear in the Literal Thinker variant C result on Opus, where two prompts with similar targets (overlooking incompleteness) produce dramatically different outcomes.

5 Probing analysis

The behavioral results establish that persona prompts selectively modulate scalar implicature comprehension. We now ask whether this modulation reflects a change in what the model represents, or in how those representations are organized. We investigate this with linear probes on the frozen activations of Qwen 3 8B.

5.1 Methods

We apply linear probing (Alain and Bengio, 2018) to the activations of Qwen 3 8B at each layer. All probes use logistic regression with five-fold cross-validation; we report mean accuracy ± 1 SD across folds.

Probe target. All probes are trained on the activation of the last token of the prompt, prior to output generation. This captures the model’s representation after processing the full input but before the response begins, so any information recoverable here reflects what was encoded during the forward pass. An initial analysis probing the final response token showed large apparent persona differences, with the Literal Thinker reaching the highest accuracy. However, this turned out to be due to response length: Literal Thinker produced shorter responses (typically just “yes”/“no”), placing the final response token closer to the prompt boundary and thus closer to the pre-decision representation. Switching to the final prompt token eliminated the apparent difference entirely. This confound is itself informative: it demonstrates that the apparent persona divergence was driven by differences in response generation rather than differences in the pre-decision representation, and motivates the use of the prompt token as a cleaner probe target throughout.

Within-persona probes (Section 5.2). The primary probe classifies underinformative vs. non-underinformative items (chance = 0.5), trained and evaluated within each persona condition separately. To characterize the layer-wise transition from floor to ceiling accuracy, we fit a sigmoid function to each persona’s accuracy curve: $\text{acc}(l) = a + (b - a)/(1 + \exp(-k(l - l_0)))$, where l_0 is the inflection point, k is the steepness, and a and b are the floor and ceiling parameters. A three-way probe (true vs. false vs. underinformative; chance = 0.33) serves as a confirmatory analysis.

Conjunctive vs. quantifier probes (Section 5.3). To test whether the two implicature subtypes are represented distinctly, we train two separate binary probes: one classifying underinformative-conjunctive vs. all other items, and one classifying underinformative-quantifier vs. all other items (chance = 0.5 in both cases). The setup is otherwise identical to the within-persona probe.

Transfer probe (Section 5.4). To test whether personas alter the geometric organization of representations even when information content is preserved, we train a probe on baseline activations only and evaluate it on persona-conditioned activations without retraining. If transfer accuracy matches baseline cross-validated accuracy, the persona preserves the baseline’s geometry; if it is degraded, this is evidence for reorganization of information. To summarize layer-wise results, we define three analysis windows based on the sigmoid fits from the within-persona probes: an early window (layers 0–15) covering the pre-inflection floor, a middle window (layers 19–25) covering the post-inflection plateau, and a late window (layers 30–35) covering the output preparation regime identified by Liu et al. (2024). Window boundaries are derived from the baseline sigmoid: all persona inflection points fall at $l_0 \geq 16.7$ (Table 3), so layer 15 is unambiguously pre-inflection across all conditions. For each window, we compute the mean transfer accuracy across layers within the window for each cross-validation fold, yielding five independent observations per persona per window. We report one-sample t -tests against chance (0.50). Charts for layers 16–18 and 26–29 can be found in Appendix C.

5.2 Probe accuracy is consistent across personas

Figure 8 shows probe accuracy by layer for all five persona conditions. At early layers (0–15), accuracy hovers between 0.65 and 0.75, with some variability both across layers and across folds. A sharp increase occurs around layers 15–19, after which accuracy reaches near-ceiling levels (≥ 0.98) and remains there through the final layer. This pattern is consistent across all five conditions. While individual persona means occasionally fall outside the baseline ± 1 SD band (11 layer–persona pairs out of 144), these deviations are small (maximum $\sim 7\text{pp}$), unsystematic in direction, and concentrated in the early layers where variance across folds is highest and near the ceiling where single-example errors create apparent difference as the standard deviation for baseline is 0.

Table 3 reports the sigmoid fit parameters. All fits achieve $R^2 > 0.98$, confirming that the sigmoid captures the shape of the transition well. Inflection points fall within a 0.7-layer range ($l_0 = 16.7\text{--}17.4$), and floor accuracies are within 3 percentage points of each other (70.1%–73.0%). Ceiling parameters are effectively identical across conditions ($\approx 100\%$). Steepness varies somewhat more across personas ($k = 0.94\text{--}1.69$), suggesting possible differences in how abruptly the representation is crystallized, though with only five folds per condition this variation may reflect fitting noise rather than a meaningful difference.

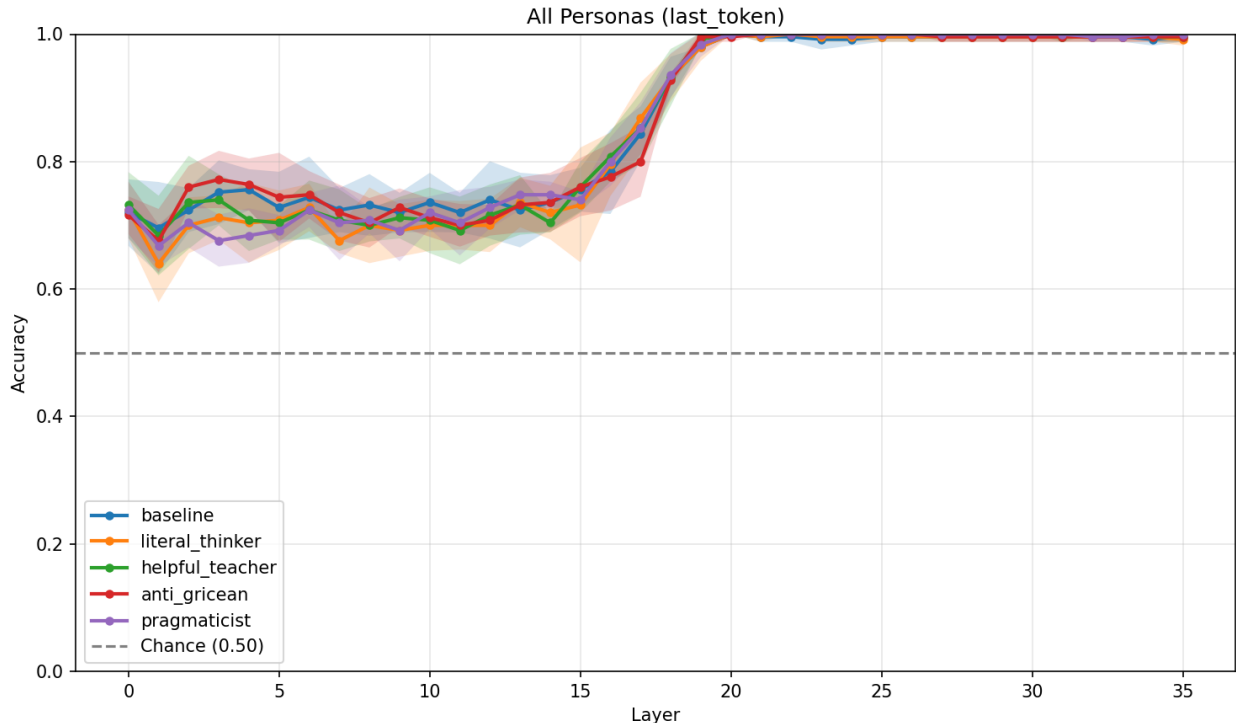


Figure 8: Linear probe accuracy by layer for underinformative vs. non-underinformative classification across persona conditions. All probes use five-fold cross-validation; shaded regions show ± 1 standard deviation across folds.

Given that the information being tested here is semantic/pragmatic, this overall pattern—ceiling performance in later layers—is broadly consistent with the layered processing hierarchy reported in prior probing work. Tenney et al. (2019) find that classical NLP pipeline stages more associated with semantics and pragmatics (semantic role labeling, coreference resolution) are recoverable at later layers in BERT than those associated with syntax (POS tagging, parsing). Jawahar et al. (2019) similarly identify a syntax→syntax→semantics gradient. However, these results are from encoder models with bidirectional attention, where every token attends to the full sequence at every layer. In decoder-only models, attention is unidirectional: each token only sees leftward context, and the final token must progressively integrate the entire sequence into a representation suitable for next-token prediction. This architectural difference may affect the layer-wise hierarchy. Liu et al. (2024) find that in Llama 2, lexical semantic information peaks in early layers while prediction peaks in later layers. The pragmatic distinction probed here reaching a ceiling only in later layers is consistent with pragmatic information being computed after and being built on the lexical and compositional semantics have been established in earlier layers. However, probing the final prompt token conflates pure representation with preparation for the upcoming decision (Liu et al., 2024), so we cannot cleanly attribute this to a dedicated “pragmatic processing” stage.

The three-way probe (true vs. false vs. underinformative; chance = 0.33) confirms the same trajectory, reaching ceiling accuracy by layer 19 with no persona-dependent differences (see Appendix B).

The central finding is that persona prompts, despite producing large behavioral differences in scalar implicature comprehension (Section 4.2), do not produce detectable differences in information content. The information needed to distinguish underinformative from non-underinformative items is represented equivalently regardless of whether the model has been instructed to accept or reject underinformative statements. At the level of information content, this is consistent with preserved pragmatic knowledge across personas. Whether personas also reorganize that information or if they only shift a decision boundary is addressed in the transfer analysis below.

Persona	Inflection (l_0)	Steepness (k)	Floor (%)	R^2
Baseline	17.1	1.30	73.0	0.994
Anti-Gricean	17.4	1.69	73.0	0.981
Helpful Teacher	16.8	1.08	71.3	0.992
Literal Thinker	16.7	1.08	70.1	0.990
Pragmaticist	16.8	0.94	70.5	0.990

Table 3: Sigmoid fit parameters for layer-wise probe accuracy across persona conditions (Qwen 3 8B).

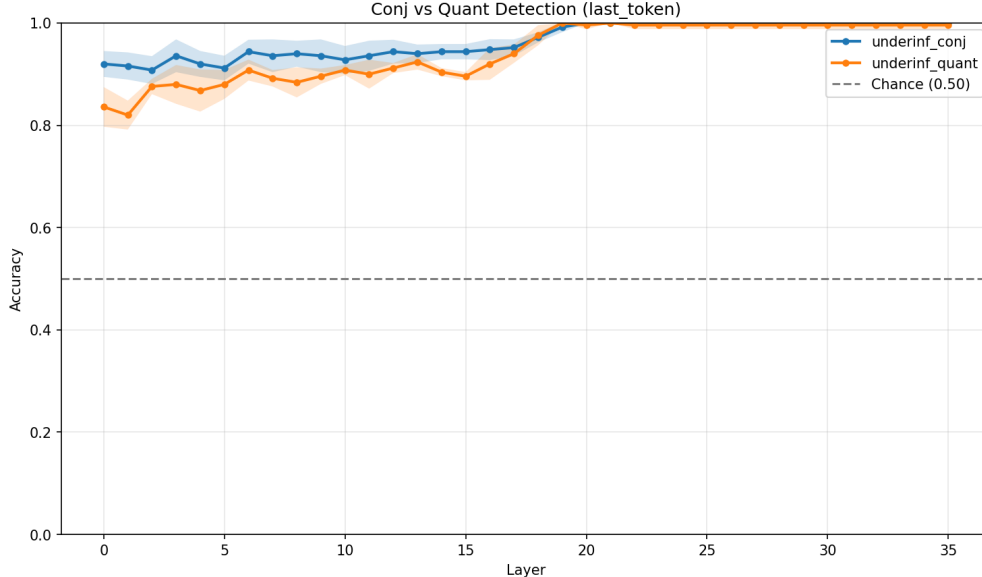


Figure 9: Linear probe accuracy by layer for underinformative-conjunctive vs. other (blue) and underinformative-quantifier vs. other (orange), baseline condition. Shaded regions show ± 1 SD across five folds.

5.3 Distinct representations for conjunctions and quantifiers

The preceding analysis pools all underinformative items into a single category. However, the behavioral results (Section 4.2) show that personas affect conjunctive and quantifier implicatures asymmetrically—for instance, the Anti-Gricean prompt nearly eliminates conjunctive implicature in Qwen while leaving quantifier implicature largely intact. This raises the question of whether the two implicature types are also represented differently in the model’s internal states.

For the conjunctive probe, accuracy is consistent across persona conditions (cross-persona range: 1.6–2.4pp; Friedman $p = 0.07$). For the quantifier probe, a Friedman test reveals a small but significant persona effect in early layers ($p < 0.001$), with a monotonic ordering: Baseline achieves the highest early-layer accuracy (+2.2pp above grand mean) and Pragmaticist the lowest (−2.6pp). These deviations are small in absolute terms (<3pp) and are dwarfed by the conjunctive–quantifier gap at the same layers (~5–7pp). At convergence (layer 19+), cross-persona range collapses to <1.6pp for both probes. We plot the baseline condition in Figure 9 as representative, though the slight difference in quantifier cases suggests that the equivalence found in Section 5.2 may be equivalent recoverability from different internal representations and not identical representations.

Figure 9 shows the layer-wise accuracy of both probes. The overall trajectory is similar: both follow the same floor-to-ceiling transition observed in the pooled probe, reaching near-perfect accuracy by approximately layer 19. However, the two probes diverge in the intermediate layers. In layers 0–17, the conjunctive

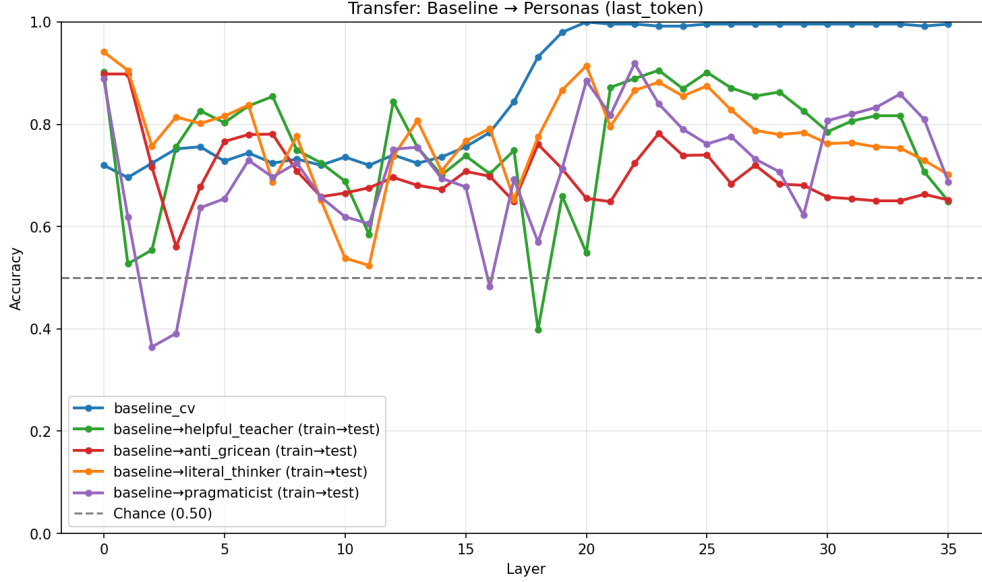


Figure 10: Transfer probe accuracy by layer: probe trained on baseline activations, tested on personas.

probe achieves consistently higher accuracy (~ 0.92 – 0.95) than the quantifier probe (~ 0.83 – 0.88), with ± 1 SD bands that do not overlap across most of this range. After layer 18, the gap closes and both probes converge to ceiling.

This separation indicates that the model does not treat conjunctive and quantifier implicature as a single representational category. The conjunctive distinction—whether the statement mentions all of the listed items—is more linearly separable from the model’s intermediate representations than the quantifier distinction—whether the statement uses an appropriate quantifier. One possible explanation is that conjunction-based implicature reduces to something closer to set-membership verification (checking whether all mentioned items appear in the statement), which can be resolved from relatively shallow lexical matching, while quantifier-based implicature requires reasoning about proportions and scale relationships (*some* vs. *all*), a computation that takes longer to crystallize into a linearly decodable form.

This representational asymmetry is consistent with the behavioral dissociations observed in Section 4.2. If conjunctive and quantifier implicatures are encoded distinctly—as the intermediate-layer probe gap suggests—then it is unsurprising that the same persona prompt would not affect both types equally. Anti-Gricean’s selective suppression of conjunctive but not quantifier implicature in Qwen, and the reversed pattern in other model–prompt combinations, is compatible with persona conditioning intervening on representations that are already structurally distinct. We note, however, that though the probe results establish representational distinctness, they do not represent a directional prediction about which type should be more susceptible to which prompt: the behavioral asymmetries are consistent with the probe findings but not explained by them.

5.4 Transfer probe

The probes in Sections 5.2 and 5.3 demonstrate that the underinformative distinction is recoverable under all persona conditions, but potentially not identically represented. A probe trained within a given persona condition finds whatever linear decision boundary works for that condition’s geometry; if personas reorganize the representational space while preserving information content, underlying structure could change even if within-persona probes showed no difference.

Figure 10 shows transfer accuracy by layer for all four persona conditions. Unlike the within-persona probes, which converge to near-identical ceiling performance by layer 19, the transfer probes show substantial persona-dependent variation across the full depth of the network. The layer-by-layer curves are much noisier than the within-persona case (which is expected since domain shift introduces noise), but three findings

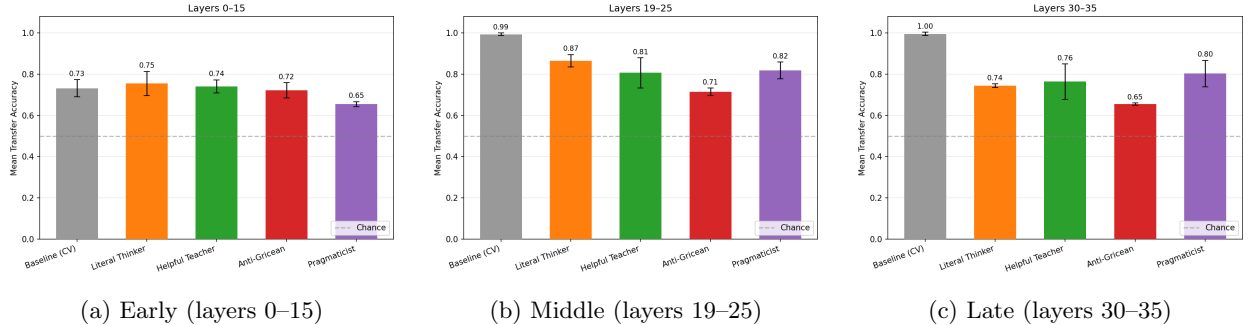


Figure 11: Mean transfer probe accuracy by window. Bars show fold-level means ($n = 5$); error bars show ± 1 SD. Dashed line indicates chance (0.50). Gray bar is baseline cross-validated accuracy (upper bound). Same color scheme as preceding figures.

emerge from the windowed analysis (Figure 11; Table 4).

Transfer is above chance but never reaches baseline. All personas achieve significantly above-chance transfer in all three windows ($p < .01$; Table 4). This confirms that the baseline probe recovers meaningful structure from persona-conditioned representations: the underinformative distinction is not just present (Section 5.2) but is encoded in a partially overlapping geometry across conditions. However, no persona approaches baseline cross-validated accuracy in any window. Even at peak—the middle window, where within-persona probes are at ceiling—the best-transferring persona (Literal Thinker, $M = 0.87$) falls 12 percentage points below baseline ($M = 0.99$). In the line graph, we see some individual layers outperform the baseline (Figure 10) but this is not reflected in the windowed results (Table 4). The spikes reflect expected sampling variability. The gap between within-persona recoverability (≥ 0.98) and cross-persona transfer (0.71–0.87) demonstrates that the same information is encoded in a different geometric configuration under persona conditioning.

Anti-Gricean shows persistent geometric displacement. The Anti-Gricean persona achieves the lowest transfer accuracy in every window: 0.72 (early), 0.71 (middle), 0.65 (late). The middle-window result is particularly striking: Anti-Gricean’s transfer accuracy is barely above its early-layer floor, even when within-persona accuracy reaches its ceiling. The other personas show a clear increase in transfer accuracy from early to middle windows (Literal Thinker: +12pp; Helpful Teacher: +7pp; Pragmaticist: +17pp), whereas Anti-Gricean remains essentially flat. This suggests that Anti-Gricean systematically reorganizes its representational geometry which persists across the full depth of the network.

The declining late-layer trajectory (0.71 $\rightarrow$ 0.65) indicates that this displacement worsens during output preparation, consistent with the persona exerting additional geometric pressure as the model prepares its response.

Other personas show disruption then partial recovery. The remaining three personas show a different trajectory: moderate transfer in early layers, peak transfer in the middle window, and partial degradation in late layers. Pragmaticist shows the most pronounced version of this pattern, rising from the lowest early-window accuracy ($M = 0.65$) to match Helpful Teacher in the middle window (0.82 vs. 0.81), and achieving the highest transfer late-window accuracy (0.80). Literal Thinker shows the highest middle-window transfer (0.87) but declines to 0.74 in late layers, comparable to Helpful Teacher (0.76).

The middle-window peak for these personas coincides with the layers where within-persona probes reach ceiling, suggesting that when the pragmatic computation crystallizes, it does so in a geometry that partially, but not fully, overlaps with baseline. The late-layer degradation across all personas is consistent with persona-specific output preparation which widens the gap between persona and baseline representations.

Window	Condition	M	SD	$t(4)$	p
Early (0–15)	Baseline (CV)	0.73	0.042	12.45	<.001
	Literal Thinker	0.75	0.059	9.66	<.001
	Helpful Teacher	0.74	0.032	16.84	<.001
	Anti-Gricean	0.72	0.037	13.28	<.001
	Pragmaticist	0.65	0.013	27.07	<.001
Middle (19–25)	Baseline (CV)	0.99	0.006	176.16	<.001
	Literal Thinker	0.87	0.030	27.38	<.001
	Helpful Teacher	0.81	0.073	9.39	<.001
	Anti-Gricean	0.71	0.018	26.01	<.001
	Pragmaticist	0.82	0.040	17.70	<.001
Late (30–35)	Baseline (CV)	1.00	0.009	127.42	<.001
	Literal Thinker	0.74	0.009	61.36	<.001
	Helpful Teacher	0.76	0.086	6.88	.002
	Anti-Gricean	0.65	0.006	57.11	<.001
	Pragmaticist	0.80	0.065	10.48	<.001

Table 4: Transfer probe accuracy by window and persona condition. Each observation is the fold-level mean across layers within the window ($n = 5$ folds). All conditions are significantly above chance (0.50).

Summary

The transfer probe reveals that persona prompts alter the representational geometry in which that information is encoded, despite preserving the information content required to distinguish underinformativeness (Section 5.2). This dissociation between information content, which is the same, and representational geometry of the information content, which is different, holds throughout the network. However, this differs between layers of the network and between personas.

The pattern supports a model in which personas do not simply adjust the decision threshold based on the same internal representation, but reorganize the representational space through which the model generates its output. The within-persona probes and the transfer probes suggest that though the same information is present, it is structured differently, which mediates shared representations but divergent behavior. To further examine this distinction, we consider the computational pathways by which the model derives its output in Section 6.

6 Attention analysis

The probing results (Section 5) establish that persona prompts preserve the information content needed to distinguish underinformative items but alter the representational geometry in which that information is encoded. We now examine the computational pathways through which the model processes the task, using attention patterns as a window into which parts of the input the model consults at different stages of processing.

6.1 Methods

We extract attention weights from all 36 layers and 32 heads of Qwen 3 8B across all five persona conditions and six item categories. We analyze three directional attention patterns, each corresponding to a distinct computational role:

- **Last token $\rightarrow$ outcome** (L $\rightarrow$ O): how much the final pre-generation position attends to the outcome region, reflecting outcome retrieval during decision preparation.
- **Last token $\rightarrow$ statement** (L $\rightarrow$ S): how much the final position attends to the statement region, reflecting statement encoding during decision preparation.
- **Statement $\rightarrow$ outcome** (S $\rightarrow$ O): how much statement tokens attend to outcome tokens, reflecting direct comparison between the two.

All values reported are raw attention mass averaged across heads within each layer. For quantitative summaries, we define three analysis windows matching Section 5.4: early (layers 0–15), middle (layers 19–25), and late (layers 30–35).

Preprocessing. Initial analysis of L $\rightarrow$ S attention revealed that sentence-initial articles (“The,” “the”) absorbed 50–70% of attention mass at early layers, consistent with the attention sink phenomenon documented in streaming language models (Xiao et al., 2024). These tokens are excluded from the content regions in all analyses.

Normalization. For L $\rightarrow$ O and L $\rightarrow$ S, where the source is a single token and the target is a content region, normalization corrects for the mechanical dilution of attention mass across a longer sequence when persona tokens are present. For S $\rightarrow$ O, where both source and target are content regions, normalization introduces a different artifact: because the numerator (attention between two content regions) is similar across conditions while the denominator (total non-persona attention) is smaller for persona conditions, normalization inflates persona conditions relative to baseline. We therefore report L $\rightarrow$ O with normalization and S $\rightarrow$ O without normalization; unnormalized L $\rightarrow$ O and normalized S $\rightarrow$ O figures are in Appendix D.

Head selection for visualizations. For single-example visualizations, we select representative attention heads automatically: for S $\rightarrow$ O, the three heads with highest mean attention mass across all source–target positions (searched across all layers); for L $\rightarrow$ O and L $\rightarrow$ S, the three heads with highest last-token-to-region mass, restricted to layers ≤ 10 and excluding degenerate heads where $>80\%$ of mass falls on a single token. Head selection is performed on the baseline condition; the same heads are then visualized across all persona conditions to ensure a fair comparison.

6.2 Last token $\rightarrow$ outcome: early-layer verification with persona-dependent dissociation

Figure 12 shows L $\rightarrow$ O attention by category under the baseline condition. Virtually all L $\rightarrow$ O activity is concentrated in layers 0–7, with attention mass dropping to near-zero by layer 10 and remaining flat through the rest of the network. This early concentration suggests that outcome retrieval is an initial encoding operation rather than a late decision-stage process.

Two category-level patterns emerge in the early window. First, conjunctive items receive substantially more L $\rightarrow$ O attention than quantifier items: false-conj leads at 0.0030, followed by underinf-conj (0.0025) and true-conj (0.0024), while all three quantifier categories cluster around 0.0011. This $\sim 2:1$ ratio is consistent across persona conditions. This likely reflects the different strategies used for verification across the two types. Conjunctive items require checking whether specific objects mentioned appear in the outcome, which benefits from attention to the outcome. Quantifier items require reasoning about the specific use of the quantifier term in the statement, which benefits from attention to the statement.

Second, within conjunctive items, the ordering is false $>$ underinf $>$ true. The model checks the outcome most when the statement is false, and least when the statement is fully correct. Underinformative items fall in between, consistent with their status of being literally true but pragmatically incomplete. This increased attention in false and underinformative cases reflects detection of a mismatch between the statement and the outcome.

Figure 13 shows L $\rightarrow$ O attention by persona, averaged across conjunctive categories. The Anti-Gricean persona produces an early-layer spike in layers 3–6. This persona effect has a directional dissociation between

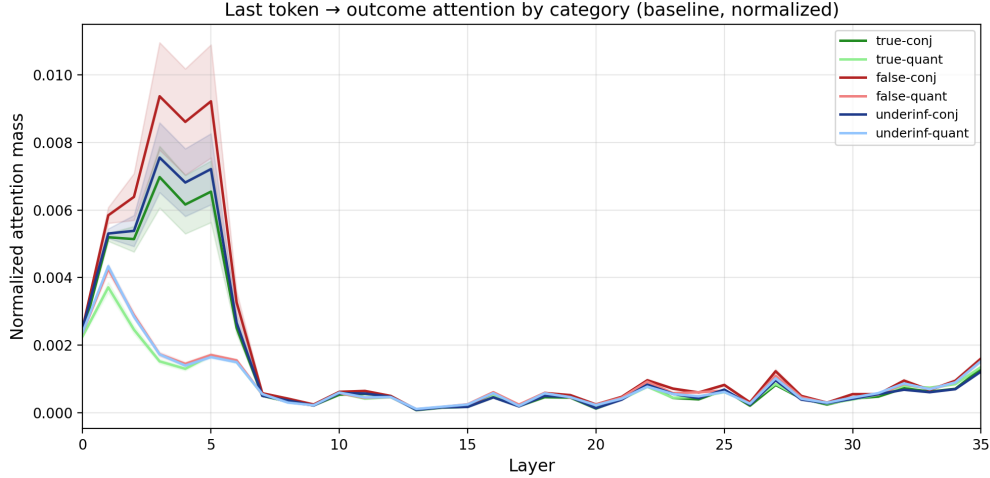


Figure 12: Last token $\rightarrow$ outcome attention by category (baseline), with ± 1 SD bars. Attention is concentrated in layers 0–7, with conjunctive items receiving approximately twice the attention mass of quantifier items.

conjunctive and quantifier cases: Anti-Gricean increases L $\rightarrow$ O attention for conjunctive items relative to baseline (+0.0006 across conjunctive items) and decreases it for quantifier items (−0.0003 across quantifier items).

This dissociation maps directly onto the behavioral results from Section 4.2, where Anti-Gricean nearly eliminates conjunctive implicature (underinf-conj: 0.98 $\rightarrow$ 0.05) while leaving quantifier implicature intact (U-quant: 0.95 $\rightarrow$ 0.93). The persona that suppresses conjunctive implicature is the same one that increases outcome-checking for conjunctive items. One interpretation is that Anti-Gricean’s heightened outcome retrieval facilitates item-by-item verification (“the biscuit is indeed in the outcome, so the statement is true”) at the expense of completeness checking (“but the statement doesn’t mention the orange”), shifting the model’s verification strategy toward literal truth and away from informativeness.

The remaining personas show lower L $\rightarrow$ O attention than baseline, with the magnitude ordered by prompt length: Pragmaticist (longest prompt) shows the largest reduction, Literal Thinker intermediate, and Helpful Teacher slightly below baseline. This likely reflects a mechanical effect: longer persona prompts dilute attention mass across more tokens, reducing the share available for any given content region. Anti-Gricean’s ability to increase the proportion of attention to outcome L $\rightarrow$ O in conjunctive cases, despite the persona adding length, makes this effect more notable.

6.3 Last token $\rightarrow$ statement: quantifier items drive statement reading

L $\rightarrow$ S attention shows a complementary pattern to L $\rightarrow$ O. Where L $\rightarrow$ O is dominated by conjunctive items, L $\rightarrow$ S is dominated by quantifier items—particularly underinformative-quantifier items, which receive dramatically more L $\rightarrow$ S attention than any other category in the early window (baseline: 0.0033 vs. next highest 0.0019 for false-quant). This pattern holds across all persona conditions.

The L $\rightarrow$ O and L $\rightarrow$ S patterns together reveal complementary verification strategies for the two implicature types. For conjunctive items, the model preferentially consults the outcome—checking whether specific items are present. For quantifier items, the model preferentially reads the statement—encoding the quantifier (*some*, *all*, or a numeral) that determines the scalar relationship. This division of labour is consistent with the hypothesis from Section 5.3 that the two implicature types involve different computations: set-membership verification for conjunctions and scalar reasoning for quantifiers.

The underinformative-quantifier spike is particularly informative. These are the items where the statement says *some* but the outcome established *all*—the classic scalar implicature case. The model devotes substantially more early attention to reading the quantifier in the statement for exactly these items, suggesting that the presence of *some* in a context where *all* would be expected triggers heightened statement

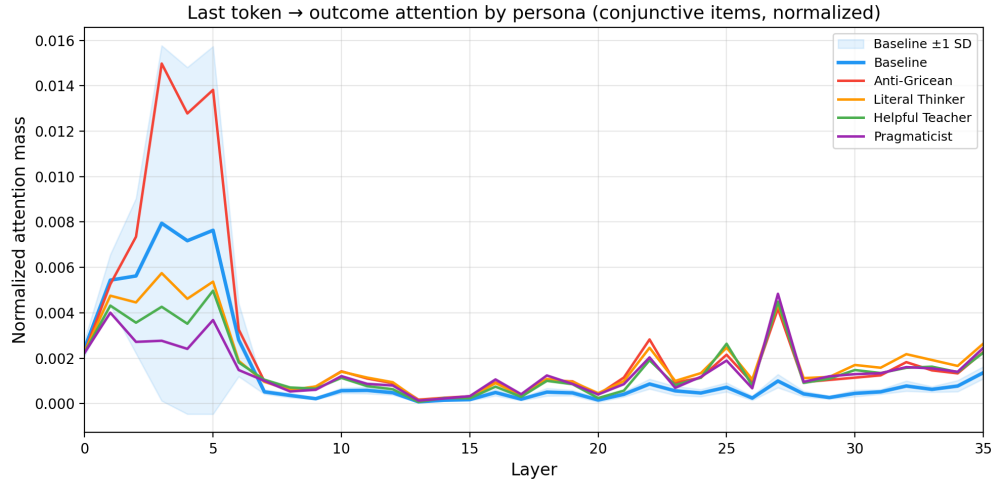


Figure 13: Last token $\rightarrow$ outcome attention by persona, averaged across conjunctive categories, with baseline ± 1 SD. The Anti-Gricean persona produces an early-layer spike at layers 3–6.

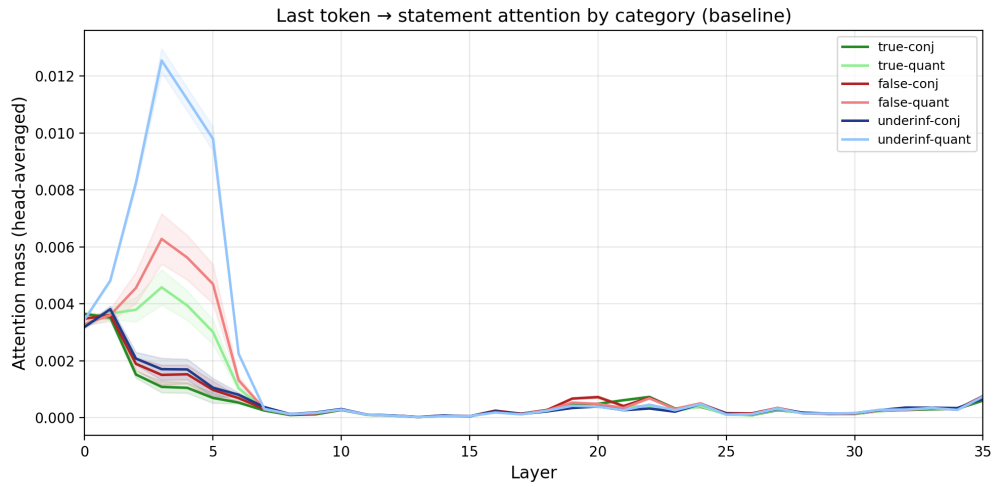


Figure 14: Last token $\rightarrow$ statement attention by category (baseline). Underinformative-quantifier items (light blue) receive substantially more statement attention than all other categories in layers 0–7.

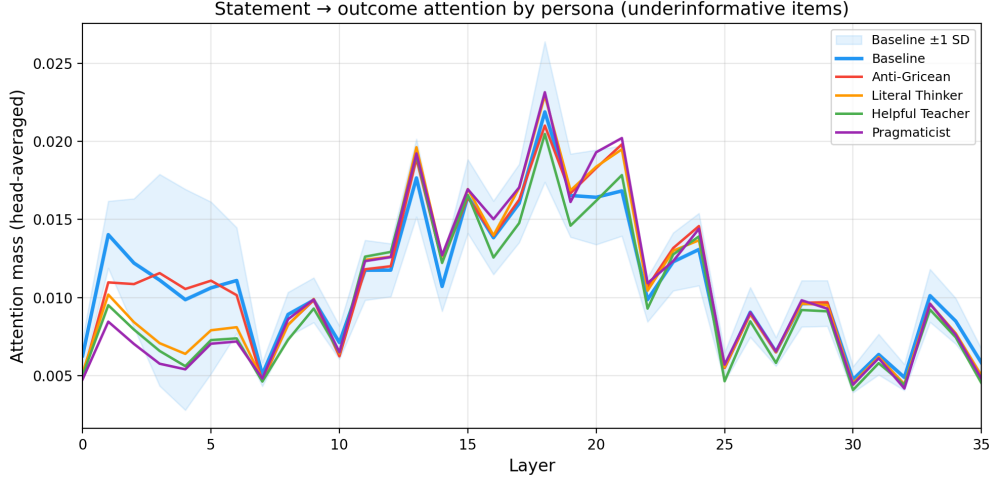


Figure 15: Statement $\rightarrow$ outcome attention by persona for underinformative items, with baseline ± 1 SD. Early-layer divergence converges by the middle window.

processing from the earliest layers.

As with $L \rightarrow O$, $L \rightarrow S$ activity is concentrated in early layers (0–7) and drops to near-uniform levels by layer 10, with all categories converging to ~ 0.0003 in the middle and late windows. The early-layer concentration of both $L \rightarrow O$ and $L \rightarrow S$ suggests that the last token’s retrieval of task-relevant content occurs during initial encoding, with later layers devoted to processing and comparison rather than further retrieval.

Persona effects on $L \rightarrow S$ are modest (see Appendix Figure 19) and largely track prompt length, with longer prompts producing lower raw attention mass to content regions. Pragmaticist shows the strongest reduction (underinf-quant early: 0.0011 vs. baseline 0.0033), though this is likely a dilution effect as the Pragmaticist prompt is the longest, rather than a meaningful reduction in statement processing.

6.4 Statement $\rightarrow$ outcome: comparison converges across personas

Figure 15 shows raw (unnormalized) $S \rightarrow O$ attention by persona for underinformative items. In contrast to $L \rightarrow O$ and $L \rightarrow S$, which are concentrated in early layers, $S \rightarrow O$ attention builds through the network, rising from ~ 0.008 in early layers to a peak of ~ 0.014 in the middle window (layers 13–20) before declining in late layers. This later temporal profile is consistent with a comparison operation that depends on prior encoding: the model first retrieves and processes the statement and outcome (via $L \rightarrow S$ and $L \rightarrow O$ in early layers), then compares them (via $S \rightarrow O$ in middle layers).

In early layers (0–7), persona conditions show modest divergence from the baseline ± 1 SD band. Baseline and Anti-Gricean show slightly elevated $S \rightarrow O$ attention (early underinf-conj: 0.0117 and 0.0118) compared to the other three personas (~ 0.0092 – 0.0096). By the middle window, all personas converge: the cross-persona range for underinf-conj narrows to 0.0130–0.0143 and for underinf-quant to 0.0125–0.0141. The comparison operation begins at different intensities across personas but arrives at the same computational endpoint.

$S \rightarrow O$ attention increases through the network, consistent with a comparison operation that builds through mid and late layers rather than occurring as a discrete early event. Figure 16 visualizes this at the token level for test item 39 (underinf-conj: “The wizard mixed the green potion” when the outcome was “the purple potion and the green potion”). At layer 5, attention flows almost exclusively from “green” in the statement to “green” in the outcome—a lexical identity check. By layer 17, the pattern changes qualitatively: statement tokens attend more broadly across outcome tokens, including tokens they do not share. This shift from lexical matching to distributed comparison is consistent with the model transitioning from surface-level token identity to something closer to propositional verification—checking not just whether individual words recur, but whether the statement as a whole is consistent with the outcome.

This is noteworthy independent of the persona manipulation. The $S \rightarrow O$ attention pattern provides a direct window into the verification computation underlying the task: the model must determine whether the

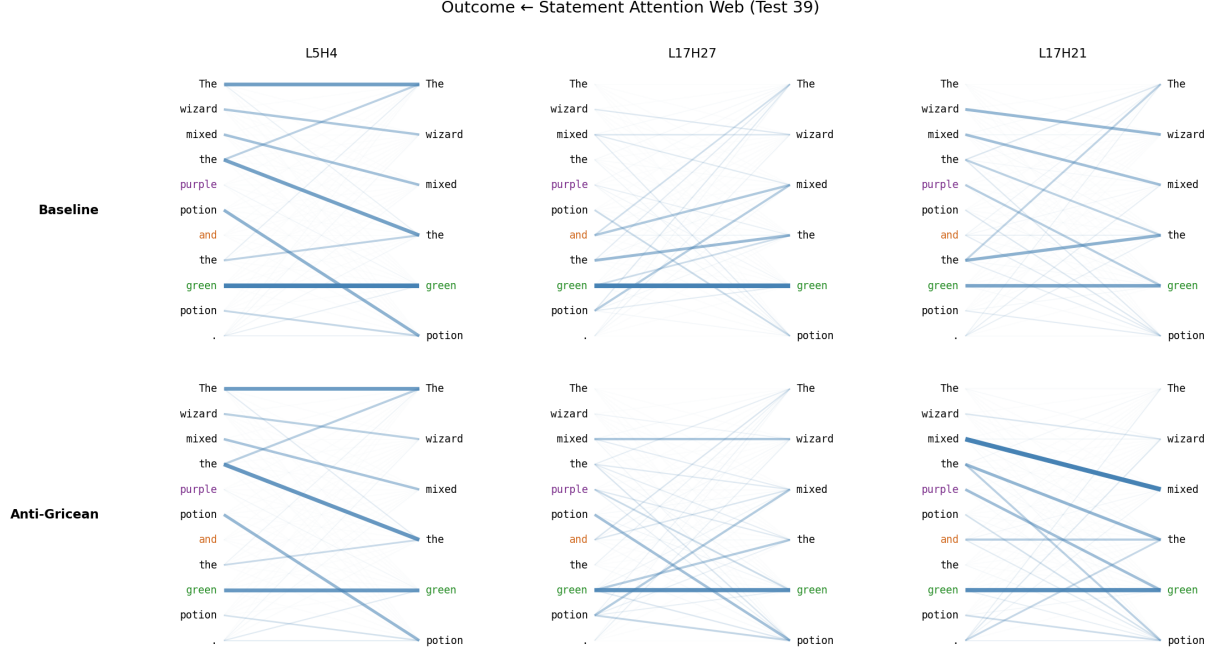


Figure 16: Statement $\rightarrow$ outcome attention for test item 39 (underinf-conj), visualized for automatically selected heads at layers 5 and 17. Baseline (top) and Anti-Gricean (bottom) show qualitatively similar comparison patterns: early lexical matching (“green” $\rightarrow$ “green”) gives way to broader compositional comparison. Full five-persona visualization in Appendix E.

statement accurately and completely describes the outcome, and the layer-wise progression from lexical to compositional comparison reflects the stages of that determination.

Summary

The attention analysis reveals complementary verification strategies with a clear temporal structure.

In early layers (0–7), the model performs retrieval via two complementary pathways: $L \rightarrow O$ attention is concentrated on conjunctive items, reflecting set-membership verification against the outcome, while $L \rightarrow S$ attention is concentrated on quantifier items, reflecting encoding of scalar terms in the statement. This division of labor is consistent with the representational asymmetry identified by the probes (Section 5.3): the two implicature types are processed through distinct computational pathways from the earliest layers.

In middle layers (13–20), the model performs statement–outcome comparison via $S \rightarrow O$ attention, which builds through the network and shows a qualitative shift from lexical matching to compositional verification. This comparison operation converges across persona conditions despite early-layer differences in onset.

Persona effects on raw attention mass are concentrated in $L \rightarrow O$, where the Anti-Gricean prompt produces a directional dissociation: increased outcome-checking for conjunctive items and decreased checking for quantifier items, mirroring the behavioral dissociation where Anti-Gricean eliminates conjunctive but not quantifier implicature. This is the most direct link between attention patterns and behavioral outcomes observed in this study, and suggests that the Anti-Gricean persona shifts the model’s verification strategy toward item-level truth checking at the expense of completeness evaluation.

Together with the probing results, the attention analysis completes a three-level dissociation. Persona prompts preserve information content (Section 5.2), alter representational geometry (Section 5.4), and modulate computational pathways—particularly the balance between outcome-checking and statement-reading strategies ($L \rightarrow O$ vs. $L \rightarrow S$). The core comparison operation ($S \rightarrow O$) remains largely shared across conditions, with behavioral divergence arising from how the products of these computations are integrated into a final decision.

7 Discussion

Our behavioral results establish that persona prompts can selectively modulate scalar implicature comprehension without degrading general language understanding. Persona prompts affect three areas of internal representation of implicature: informational content, informational organization, and computational pathways.

7.1 Three-level dissociation

The central interpretability finding is a dissociation between what information is represented, how that information is organized, and what computational pathways the model uses to act on it.

At the level of informational content, persona prompts have no detectable effect. Linear probes trained to distinguish underinformative from nonunderinformative items reach ceiling accuracy by layer 19 regardless of persona condition (Section 5.2), with sigmoid inflection points falling within a 0.7-layer range across all five conditions. The model “knows” whether the statement is underinformative under every persona, including those where it behaves as if it doesn’t. The conjunctive/quantifier probe (Section 5.3) confirms this for both implicature subtypes individually: both reach ceiling recoverability under all persona conditions, though their trajectories diverge in intermediate layers (discussed in Section 7.3).

At the level of informational organization, personas produce substantial effects. The transfer probe (Section 5.4) shows that a linear classifier trained on baseline activations degrades when evaluated on persona-conditioned activations, even at layers where within-persona probes have 100% accuracy and little variability. The same information is present but is organized differently in latent space. This degradation is not uniform: the Anti-Gricean persona shows persistent geometric displacement across all network depths, while other personas show partial recovery in middle layers before degrading again in late layers as the model prepares its output.

At the level of computational pathways, the attention analysis (Section 6) reveals that personas alter which parts of the input the model consults during processing. Most strikingly, the Anti-Gricean persona increases early-layer attention from the last token to the outcome region for conjunctive items (Figure 12) while simultaneously decreasing it for quantifier items (Appendix Figure 19), a directional dissociation that maps directly onto its behavioral pattern of eliminating conjunctive but not quantifier implicature. Meanwhile, the core comparison operation (statement→outcome attention in middle layers—Figure 15) converges across personas, suggesting that the underlying verification computation is shared even when its inputs and the model’s interpretation of its outputs are not.

The three levels are separate. The transfer probe shows that information content can stay the same even when the information is reorganized, and the attention patterns show personas can alter retrieval patterns (L→O and L→S attention) without altering the comparison computation (S→O attention). This rules out the analysis of personas mediating the decision threshold without changing the internal representations. If that were the case, the transfer probe would show perfect transfer and attention patterns would be identical across conditions. The preserved information content and comparison computation also rules out the analysis that personas cause a complete loss of internal pragmatic knowledge.

The behavioral pattern observed under implicature-inhibiting personas, where representation of underinformativeness is preserved while the model accepts the utterances, reflects what Katsos and Bishop (2011) term ‘pragmatic tolerance’ in child language development. Children do not lack sensitivity to scalar implicature: they detect underinformativeness but tolerate it, accepting utterances like *some of the balls are blue* when all the balls are blue at higher rates than adults. The same ambiguity applies here: acceptance of underinformative statements could reflect either a lack of pragmatic competence or a higher threshold for rejection. The probing results favor the tolerance account: the model represents underinformativeness under all persona conditions, but the inhibiting personas cause the model to accept utterances even if internally represented as pragmatically deficient. However, the interpretability results are more complicated than the simple threshold account. Pragmatic tolerance as framed by Katsos and Bishop (2011) establishes that detection of underinformativeness can be dissociated from the decision to accept or reject it, but the paradigm cannot determine whether the underlying representations are equivalent across tolerance levels. The probing results address this gap in LLMs. The transfer probe and attention analysis show that persona-conditioned tolerance does change representation, not merely an altered threshold: informational content is reorganized

and the computational pathways are different. The direction of the analogy also differs: children are developing toward adult-like pragmatic intolerance, while personas push models away from baseline intolerance. This analogy is not a complete mechanistic account of personas, but it places these LLM results in an existing framework for understanding how pragmatic competence and pragmatic behavior can be separate.

7.2 How personas work

The trichotomy detailed above suggests that personas restructure representations and computational pathways involved in implicature processing without erasing the information itself. Regardless of persona, the model records and understands the pragmatic status of the input (whether the statement is true, false, or underinformative) but the geometric space in which this encoding lives, and the attentional pathways through which the model arrives at it, are persona-dependent.

Our account is consistent with the persona selection model (PSM; Marks et al., 2026), which proposes that pretraining develops a repertoire of characters, and posttraining selects one (“the Assistant”). On this view, persona prompts do not teach the model new information about the task; rather, they shift which character the model enacts, and different characters process the same information through different computational strategies. The preservation of informational content across personas is predicted by PSM: the model’s underlying world knowledge (including pragmatics) is a property of the LLM, and doesn’t change with persona. What changes is the character’s disposition toward that knowledge: whether it treats underinformativeness as grounds for rejection or as irrelevant.

PSM also offers a framework for understanding why the transfer probe shows geometric displacement rather than erasure. If persona prompts reorganize the model’s processing by selecting into a different region of a pre-existing persona space, as Lu et al. (2026) demonstrate for the Assistant axis in activation space, then the same pragmatic features should be recoverable under every condition (because the features are properties of the LLM’s world model) but organized differently (because the persona conditions the geometry of how those features relate to the decision). This is exactly what the transfer probe shows.

The variant analysis (Section 4.3) provides additional evidence for the PSM framing, while also revealing its limits. Across all three models, the direction of persona effects is robust to rephrasing: implicature-inhibiting prompts suppress implicature and implicature-boosting prompts enhance it, regardless of specific wording. However, the magnitude of the effect is sensitive to how the persona is framed. A particularly clear case is the Literal Thinker variant C on Claude Opus 4.5, which frames the criterion positively (“if the speaker said something true, that meets the requirements for a good answer”) rather than as a dismissal of completeness. This variant leaves implicature largely intact (82–84%) while the original and other variants suppress it to near zero. The same persona produces dramatically different behavioral outcomes depending on whether it is framed as not caring about completeness or as positively valuing truth. From a PSM perspective, the rephrased prompt has selected an adjacent but distinct character from Opus’s repertoire of characters.

The interpretability results in Sections 5 and 6 provide evidence that persona framings change how the model solves the task, not just what answer it produces. The Anti-Gricean and Literal Thinker prompts were designed as two instantiations of the same implicature suppression but they evoke different characters, and the model implements them through qualitatively different computational pathways. The Anti-Gricean persona induces a distinctive early-layer spike in last-token $\rightarrow$ outcome attention for conjunctive items (Section 6.2), consistent with redirecting the model’s verification strategy toward item-level truth checking at the expense of completeness checking. This attentional signature is absent under the Literal Thinker persona, which suppresses implicature more uniformly without altering the balance of retrieval pathways. The transfer probe corroborates this: Anti-Gricean shows the greatest and most persistent geometric displacement from baseline, while the Literal Thinker achieves the highest middle-window transfer accuracy, preserving more of the baseline geometry while still shifting behavior. In PSM terms, the Anti-Gricean prompt evokes a character that actively reasons about truth conditions, implemented through restructuring of attention, while the Literal Thinker evokes a character that simply applies a more lenient standard, implemented more subtly.

The cross-model variation in susceptibility extends this picture, though not as a simple crossover. Qwen and Opus are both responsive to both inhibiting prompts, while GPT is largely resistant to the Literal Thinker and only moderately affected by the Anti-Gricean. More strikingly, Anti-Gricean shows model-

dependent selectivity in which implicature subtype it targets: conjunctive in Qwen, both types in Opus, quantifier in GPT. A persona prompt’s effect does not depend solely on its content, but through an interaction between the LLM’s repertoire of characters and its understanding of implicature. The cross-model difference could be due to training data, RLHF procedures, or architectural properties, but the exact source is unclear, though the pattern is consistent with PSM’s prediction that different models would implement the same personas through a different internal set of learned characters.

7.3 Conjunctive and quantifier implicature as distinct

The probe and attention results converge on the conclusion that conjunctive and quantifier scalar implicature are structurally different phenomena. They are representationally distinct from intermediate layers onward (Section 5.3), processed through complementary attentional pathways (Section 6), and differentially susceptible to persona conditioning (Section 4.2).

The representational evidence comes from the conjunctive/quantifier probes, which show that the conjunctive distinction is more linearly separable from intermediate-layer activations than the quantifier distinction, with a gap of approximately 5–7 percentage points in layers 0–17 that closes only at convergence. This suggests that the two types of implicature become decodable at different rates, consistent with the hypothesis that conjunction-based implicature involves a simpler reasoning process (set-membership verification) compared to quantifier-based implicature.

The attention analysis provides a computational account of this distinction, as the two types of implicature have distinct attention signatures. For conjunctive items, the model preferentially directs early-layer attention from the last token to the outcome region ($L \rightarrow O$), consistent with checking whether specific mentioned items appear in the outcome. For quantifier items, the model preferentially directs attention to the statement ($L \rightarrow S$), consistent with encoding the scalar term (some, all, or a numeral) that determines the relationship between statement and outcome. This computational distinction is not a persona-specific effect, but applies across different conditions.

This structural difference helps explain why personas affect implicature differently in conjunctive and quantifier cases (Section 4.2). If conjunctive and quantifier implicature are both represented and processed differently, a persona that applies pressure in one neighborhood need not affect the other.

The conjunctive/quantifier difference manifests differently in different models, as noted in Section 7.2. Anti-Gricean affects quantifier but not conjunctive items in GPT, while the reverse is true in Qwen, and there is no distinction in Opus. This suggests that the mapping between persona type and implicature subtype is model-dependent. The difference in processing between conjunctive and quantifier processing may generalize across models, but the specific details of processing depend on model architecture.

7.4 Limitations and future work

The interpretability analyses presented here establish what information is represented and where computational resources are allocated, but not what is causally necessary for the behavioral outcome. The observations made through probing and attention analysis provide some possible explanations, but they are correlational observations. Persona-dependent behavioral differences may arise from mechanisms not captured by either analysis, for example, multi-layer perceptron computations that alter the residual stream without affecting attention patterns, or from representational changes orthogonal to those captured by logistic regression. Activation patching on specific components identified in Section 6 would test whether these components are causally responsible for the observed behavioral divergence or are epiphenomenal correlates of it. Another way of causally testing persona effects would be to identify if a ‘literal’/‘pragmatic’ axis exists (similar to how Lu et al. (2026) identify an “Assistant axis”) and steer along this axis to affect model output. This could also identify other effects which the model encodes along this axis, such as changes in personality.

The finding that personas modulate pragmatic reasoning through representational reorganization rather than simple threshold adjustment raises the question of whether these effects share mechanisms with Theory of Mind (ToM) computation. Recent work using functional localization has identified subnetworks in LLMs that are causally involved in both ToM and pragmatic tasks (Tsvilodub et al., 2026), but the present study did not test whether persona conditioning recruits or disrupts these subnetworks. The dissociation between the two inhibiting prompts is particularly relevant here: the Anti-Gricean prompt explicitly invokes reasoning

about truth conditions while the Literal Thinker prompt adjusts behavioral expectations without requiring such reasoning. If these two prompt types differentially recruit ToM-relevant components, this would provide a mechanistic account of their divergent effects despite both being intended to suppress implicature.

The interpretability analysis was conducted exclusively on Qwen 3 8B, the only model in this study with accessible weights. Given the cross-model differences identified behaviorally (Section 4.2), extending the interpretability analysis to more models to identify whether representational differences generalize across models would provide valuable insight. The interpretability analysis also was not conducted on the variant prompts identified in Section 4.3 due to time constraints. Further analysis would extend the present interpretability analysis to more open-weights models and more variants of prompts. This would clarify the PSM analysis by showing how prompt phrasing and model-specific properties interact with persona effects.

Time constraints prevented rerunning the corrupted Kimi-K2.5 and GLM-5 results. Future work would extend the behavioral analysis to these models to analyze effects of MoE architectures and cultural differences between Chinese- and American-trained models.

References

- Alain, G. and Bengio, Y. (2018). Understanding intermediate layers using linear classifier probes.
- Anthropic (2025). Claude Opus 4.5 system card.
- Barner, D., Brooks, N., and Bale, A. (2011). Accessing the unsaid: The role of scalar alternatives in children’s pragmatic inference. *Cognition*, 118(1):84–93.
- Belinkov, Y. (2022). Probing Classifiers: Promises, Shortcomings, and Advances. *Computational Linguistics*, 48(1):207–219.
- Bott, L. and Noveck, I. A. (2004). Some utterances are underinformative: The onset and time course of scalar inferences. *Journal of Memory and Language*, 51(3):437–457.
- Chen, R., Arditì, A., Sleight, H., Evans, O., and Lindsey, J. (2025). Persona Vectors: Monitoring and Controlling Character Traits in Language Models.
- Cho, Y.-e. and Kim, S. m. (2024). Pragmatic inference of scalar implicature by LLMs.
- Clark, K., Khandelwal, U., Levy, O., and Manning, C. D. (2019). What Does BERT Look at? An Analysis of BERT’s Attention. In Linzen, T., Chrupała, G., Belinkov, Y., and Hupkes, D., editors, *Proceedings of the 2019 ACL Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, pages 276–286, Florence, Italy. Association for Computational Linguistics.
- Cong, Y. (2022). Pre-trained Language Models’ Interpretation of Evaluativity Implicature: Evidence from Gradable Adjectives Usage in Context. In Pyatkin, V., Fried, D., and Anthonio, T., editors, *Proceedings of the Second Workshop on Understanding Implicit and Underspecified Language*, pages 1–7, Seattle, USA. Association for Computational Linguistics.
- Elazar, Y., Ravfogel, S., Jacovi, A., and Goldberg, Y. (2021). Amnesic Probing: Behavioral Explanation with Amnesic Counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175.
- GLM-5-Team (2026). Glm-5: from vibe coding to agentic engineering.
- Google DeepMind (2025). Gemini 3 pro model card.
- Grice, H. P. (1975). *Logic and Conversation*, pages 41–58. Number 3 in Syntax and Semantics. Elsevier.
- Gupta, S., Shrivastava, V., Deshpande, A., Kalyan, A., Clark, P., Sabharwal, A., and Khot, T. (2024). Bias Runs Deep: Implicit Reasoning Biases in Persona-Assigned LLMs.
- Hewitt, J. and Liang, P. (2019). Designing and Interpreting Probes with Control Tasks. In Inui, K., Jiang, J., Ng, V., and Wan, X., editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2733–2743, Hong Kong, China. Association for Computational Linguistics.
- Hewitt, J. and Manning, C. D. (2019). A Structural Probe for Finding Syntax in Word Representations. In *Proceedings of the 2019 Conference of the North*, pages 4129–4138, Minneapolis, Minnesota. Association for Computational Linguistics.
- Horn, L. R. (1972). *On the Semantic Properties of Logical Operators in English*. PhD thesis, UCLA, Los Angeles.
- Hu, J., Floyd, S., Jouravlev, O., Fedorenko, E., and Gibson, E. (2023). A fine-grained comparison of pragmatic language understanding in humans and language models. In Rogers, A., Boyd-Graber, J., and Okazaki, N., editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4194–4213, Toronto, Canada. Association for Computational Linguistics.
- Hu, T. and Collier, N. (2024). Quantifying the Persona Effect in LLM Simulations.

- Jain, S. and Wallace, B. C. (2019). Attention is not Explanation. In Burstein, J., Doran, C., and Solorio, T., editors, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 3543–3556, Minneapolis, Minnesota. Association for Computational Linguistics.
- Jawahar, G., Sagot, B., and Seddah, D. (2019). What Does BERT Learn about the Structure of Language? In Korhonen, A., Traum, D., and Màrquez, L., editors, *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3651–3657, Florence, Italy. Association for Computational Linguistics.
- Katsos, N. and Bishop, D. V. (2011). Pragmatic tolerance: Implications for the acquisition of informativeness and implicature. *Cognition*, 120(1):67–81.
- Li, K., Liu, T., Bashkansky, N., Bau, D., Viégas, F., Pfister, H., and Wattenberg, M. (2024). Measuring and Controlling Persona Drift in Language Model Dialogs.
- Liu, Z., Kong, C., Liu, Y., and Sun, M. (2024). Fantastic Semantics and Where to Find Them: Investigating Which Layers of Generative LLMs Reflect Lexical Semantics.
- Lu, C., Gallagher, J., Michala, J., Fish, K., and Lindsey, J. (2026). The Assistant Axis: Situating and Stabilizing the Default Persona of Language Models.
- Ma, B., Li, Y., Zhou, W., Gong, Z., Liu, Y. J., Jasinskaja, K., Friedrich, A., Hirschberg, J., Kreuter, F., and Plank, B. (2025). Pragmatics in the Era of Large Language Models: A Survey on Datasets, Evaluation, Opportunities and Challenges. In Che, W., Nabende, J., Shutova, E., and Pilehvar, M. T., editors, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8679–8696, Vienna, Austria. Association for Computational Linguistics.
- Manning, C. D., Clark, K., Hewitt, J., Khandelwal, U., and Levy, O. (2020). Emergent linguistic structure in artificial neural networks trained by self-supervision. *Proceedings of the National Academy of Sciences*, 117(48):30046–30054.
- Marks, S., Lindsey, J., and Olah, C. (2026). The persona selection model: Why ai assistants might behave like humans. <https://alignment.anthropic.com/2026/psm/>.
- OpenAI (2025). Update to GPT-5 system card: GPT-5.2.
- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askell, A., Welinder, P., Christiano, P., Leike, J., and Lowe, R. (2022). Training language models to follow instructions with human feedback.
- Ravichander, A., Belinkov, Y., and Hovy, E. (2021). Probing the Probing Paradigm: Does Probing Accuracy Entail Task Relevance? In Merlo, P., Tiedemann, J., and Tsarfaty, R., editors, *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 3363–3377, Online. Association for Computational Linguistics.
- Ruis, L., Khan, A., Biderman, S., Hooker, S., Rocktäschel, T., and Grefenstette, E. (2023). The Goldilocks of Pragmatic Understanding: Fine-Tuning Strategy Matters for Implicature Resolution by LLMs.
- Salewski, L., Alaniz, S., Rio-Torto, I., Schulz, E., and Akata, Z. (2023). In-Context Impersonation Reveals Large Language Models’ Strengths and Biases.
- Schulhoff, S., Ilie, M., Balepur, N., Kahadze, K., Liu, A., Si, C., Li, Y., Gupta, A., Han, H., Schulhoff, S., Dulepet, P. S., Vidyadhara, S., Ki, D., Agrawal, S., Pham, C., Kroiz, G., Li, F., Tao, H., Srivastava, A., Costa, H. D., Gupta, S., Rogers, M. L., Goncarenco, I., Sarli, G., Galynker, I., Peskoff, D., Carpuat, M., White, J., Anadkat, S., Hoyle, A., and Resnik, P. (2025). The Prompt Report: A Systematic Survey of Prompt Engineering Techniques.

- Serrano, S. and Smith, N. A. (2019). Is Attention Interpretable? In Korhonen, A., Traum, D., and Màrquez, L., editors, *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 2931–2951, Florence, Italy. Association for Computational Linguistics.
- Sravanthi, S., Doshi, M., Tankala, P., Murthy, R., Dabre, R., and Bhattacharyya, P. (2024). PUB: A Pragmatics Understanding Benchmark for Assessing LLMs’ Pragmatics Capabilities. In Ku, L.-W., Martins, A., and Srikumar, V., editors, *Findings of the Association for Computational Linguistics: ACL 2024*, pages 12075–12097, Bangkok, Thailand. Association for Computational Linguistics.
- Team, K. (2026). Kimi k2.5: Visual agentic intelligence.
- Tenney, I., Das, D., and Pavlick, E. (2019). BERT Rediscovered the Classical NLP Pipeline. In Korhonen, A., Traum, D., and Màrquez, L., editors, *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4593–4601, Florence, Italy. Association for Computational Linguistics.
- Tsvilodub, P., Klumpp, J.-F., Mohammadpour, A., Hu, J., and Franke, M. (2026). On Emergent Social World Models – Evidence for Functional Integration of Theory of Mind and Pragmatic Reasoning in Language Models.
- Tsvilodub, P., Marty, P., Ramotowska, S., Romoli, J., and Franke, M. (2024). Experimental Pragmatics with Machines: Testing LLM Predictions for the Inferences of Plain and Embedded Disjunctions.
- Xiao, G., Tian, Y., Chen, B., Han, S., and Lewis, M. (2024). Efficient Streaming Language Models with Attention Sinks.
- Yang, A., Li, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Gao, C., Huang, C., Lv, C., Zheng, C., Liu, D., Zhou, F., Huang, F., Hu, F., Ge, H., Wei, H., Lin, H., Tang, J., Yang, J., Tu, J., Zhang, J., Yang, J., Yang, J., Zhou, J., Zhou, J., Lin, J., Dang, K., Bao, K., Yang, K., Yu, L., Deng, L., Li, M., Xue, M., Li, M., Zhang, P., Wang, P., Zhu, Q., Men, R., Gao, R., Liu, S., Luo, S., Li, T., Tang, T., Yin, W., Ren, X., Wang, X., Zhang, X., Ren, X., Fan, Y., Su, Y., Zhang, Y., Zhang, Y., Wan, Y., Liu, Y., Wang, Z., Cui, Z., Zhang, Z., Zhou, Z., and Qiu, Z. (2025). Qwen3 Technical Report.
- Zheng, M., Pei, J., Logeswaran, L., Lee, M., and Jurgens, D. (2024). When "A Helpful Assistant" Is Not Really Helpful: Personas in System Prompts Do Not Improve Performances of Large Language Models.

A Full variant information

This prompt was used with Gemini 3 (Google DeepMind, 2025) to generate three new variant prompts for each persona based on the existing four personas.

I have four persona prompts used in a psycholinguistics experiment. I need paraphrase variants of each that preserve the core instruction and register while changing surface wording. For each prompt, generate 3 variants.

Constraints:

Keep the same level of formality/technicality as the original

If the original uses specific technical terms (e.g., "truth value," "maxim of quantity"), the paraphrase should use those same terms or equally technical synonyms --- don't casualize a formal prompt or vice versa

Preserve the overall framing: if the original frames the task as evaluation criteria, the variant should too; if it describes a behavioral disposition, keep that

Keep roughly the same length ($\pm 20\%$)

Don't introduce concepts absent from the original (e.g., don't add "Grice" to a prompt that didn't mention Grice)

[Original prompts provided here; see Table 2 for full text.]

The generated variants are as follows.

Anti-Gricean variants

Variant A. Your evaluation of statements depends exclusively on truth value. A response is considered adequate if and only if it is literally accurate. It is irrelevant whether the response is exhaustive, informative, or includes all events, provided it is not false.

Variant B. You judge statements based strictly on their literal truth. An answer is valid provided it contains no falsehoods; whether it provides a complete or informative account is immaterial. Partial truths satisfy the criteria for being true.

Variant C. Focus solely on the truth value of a statement when assessing it. A good answer is defined by its literal veracity—it need not be comprehensive, informative, or mention every detail. If the content is true, it remains true even if incomplete.

Helpful Teacher variants

Variant A. You act as a teacher instructing young children on how to provide quality responses. A high-quality answer must be exhaustive and convey the entire narrative rather than a fraction. You prioritize the inclusion of all essential information.

Variant B. As a teacher, you are guiding children toward providing effective answers. Good responses are complete and describe the whole story from start to finish. In your evaluation, you focus on whether the speaker provided every important detail.

Variant C. You are in the role of a teacher helping children learn to communicate well. An answer is only good if it is thorough and tells the full story. When you look at these answers, you care most about whether the child has shared all the key information.

Literal Thinker variants

Variant A. In your assessment of answers, prioritize whether the statement is factually correct. Do not be overly concerned with whether every specific detail is included—so long as the utterance is true, that is sufficient.

Variant B. When you evaluate responses, look only at whether they are factually accurate. It isn't necessary for a statement to capture the entire picture; if what was stated is fundamentally true, you should consider it acceptable.

Variant C. Focus your evaluation on the factual accuracy of the given statement. Avoid worrying about whether the answer is missing certain details—if the speaker said something true, that meets the requirements for a good answer.

Pragmaticist variants

Variant A. You are a specialist in pragmatics and Gricean communication. You assess statements based on their informativeness—not merely literal truth, but whether they provide the necessary information for the context. Omitting relevant data violates the maxim of quantity and renders the answer insufficient.

Variant B. As an expert in the field of pragmatics, you judge utterances by their communicative appropriateness. You evaluate if they offer the correct amount of information for the current context. Any answer that fails to include pertinent facts violates the maxim of quantity and is inadequate.

Variant C. You approach evaluation as an expert in Gricean communication and pragmatics. Statements are judged on being appropriately informative rather than just true. If a response leaves out relevant information, it violates the maxim of quantity and should be marked as a poor answer.

B Three-way probe results

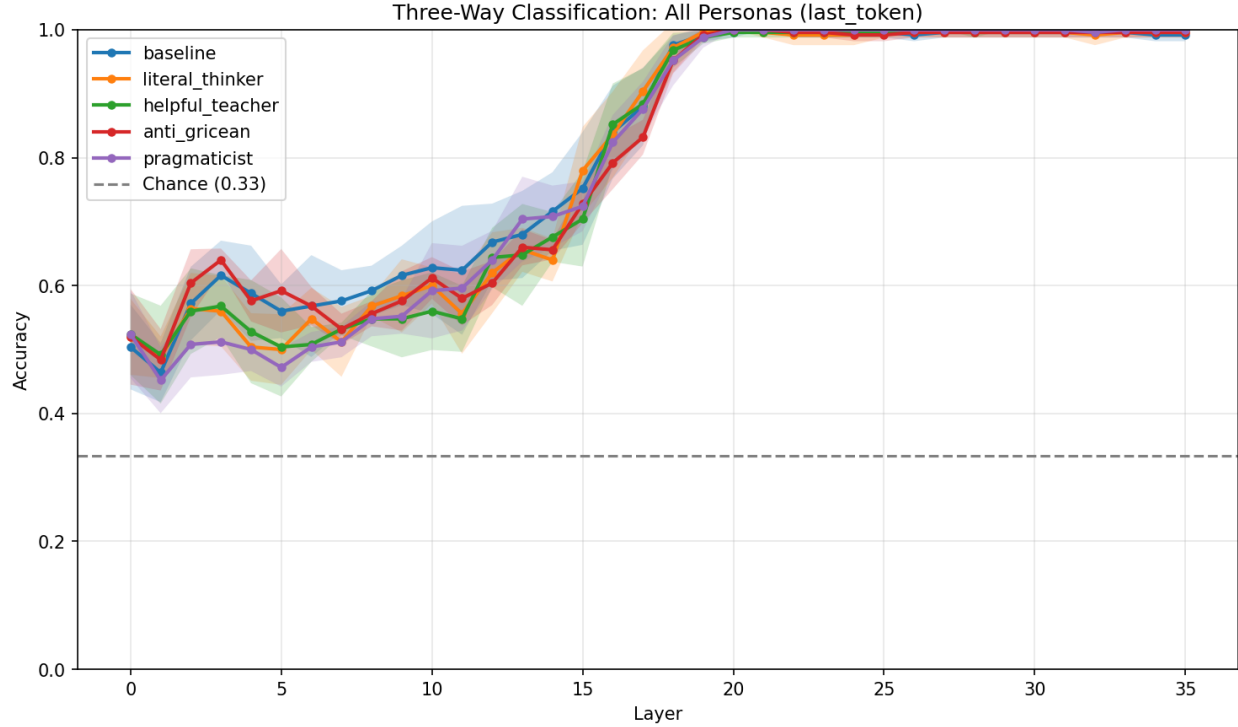


Figure 17: Three-way probe (true/false/underinformative) accuracy across layers for all persona conditions, with mean and ± 1 SD interval plotted. Chance = 0.33 (dashed line).

C More transfer layers

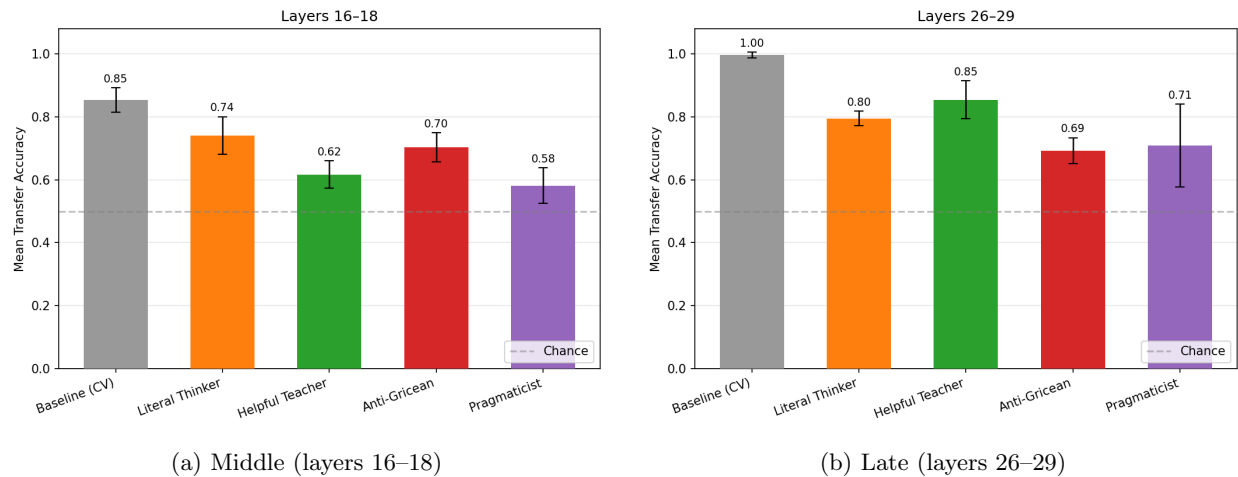


Figure 18: Mean transfer probe accuracy by window. Bars show fold-level means ($n = 5$); error bars show ± 1 SD. Dashed line indicates chance (0.50). Gray bar is baseline cross-validated accuracy (upper bound). Same color scheme as preceding figures.

D Attention analysis variants

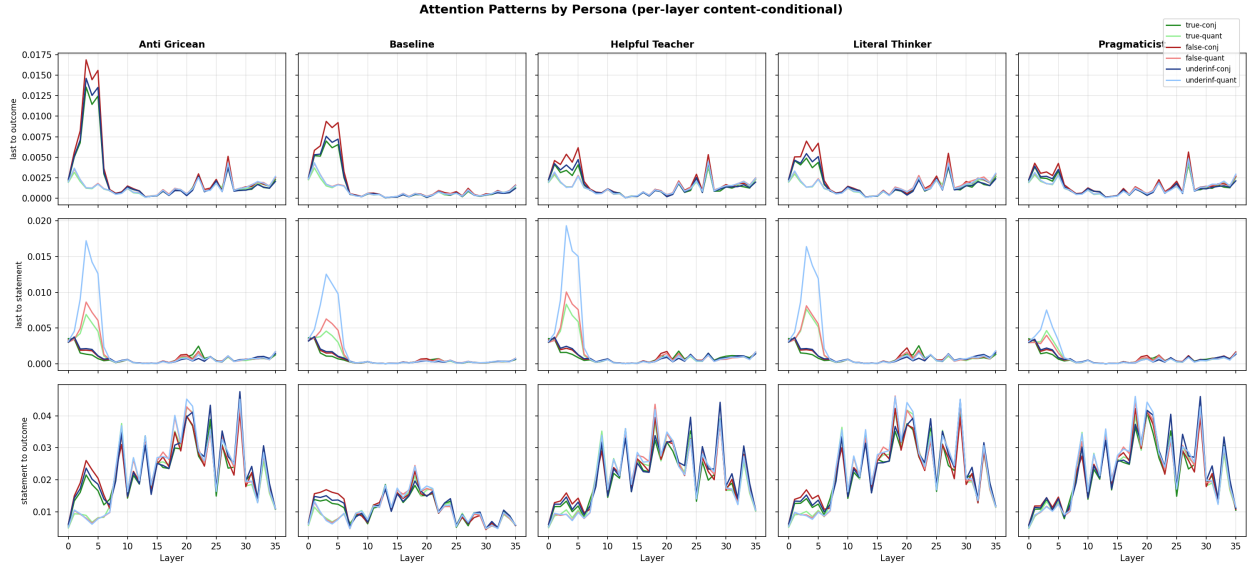


Figure 19: Average attention by question category for last token to statement, last token to outcome, statement to outcome cases. Articles are included. Attention is normalized based on share of non-persona attention by layer.

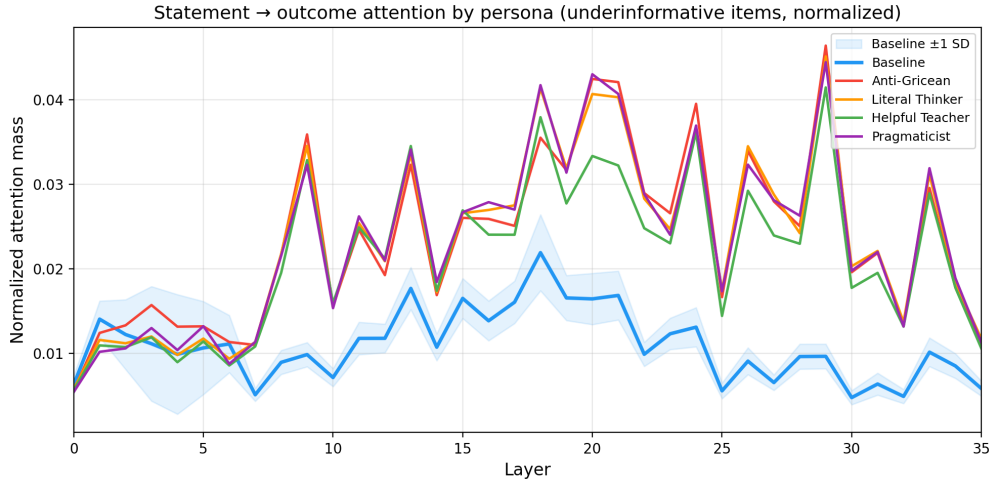


Figure 20: Statement $\rightarrow$ Outcome attention by persona, normalized by proportion of attention to content by layer. Shaded region represents baseline ± 1 SD.

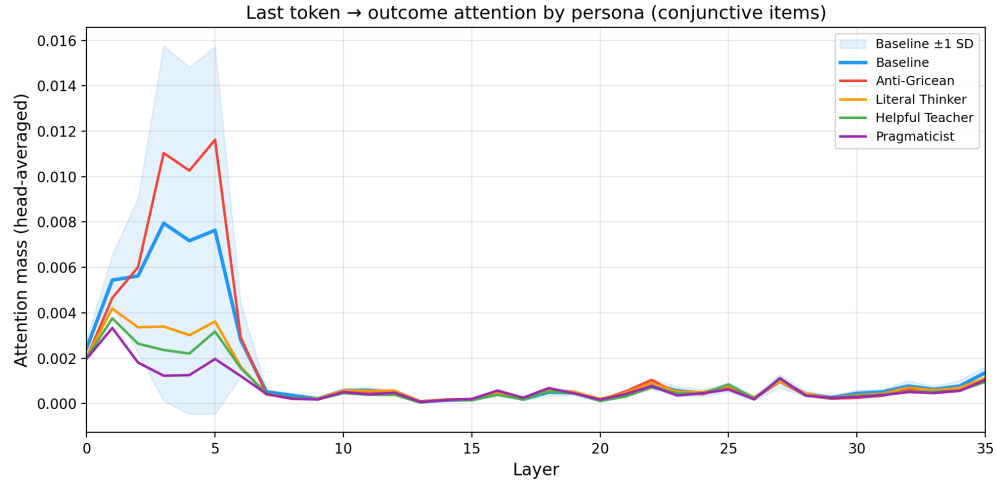


Figure 21: Last token $\rightarrow$ outcome attention by persona (conjunctive cases), with ± 1 SD bars, without normalization.

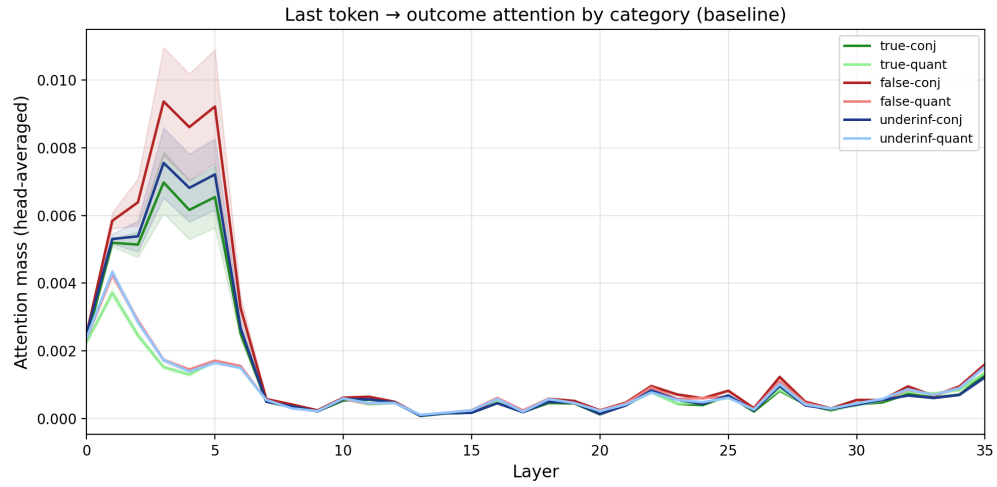


Figure 22: Last token $\rightarrow$ outcome attention by category (baseline), with ± 1 SD bars, without normalization.

E Test 39 attention visualization, all personas

Outcome ← Statement Attention Web (Test 39)



Figure 23: Visualization of three most active attention heads on example 39 for all personas and baseline condition.